

Labs 08-11: Azure Monitor, Microsoft Defender for cloud, Enable Just-In Time Access in VMs, Microsoft Sentinel

Introduction

In today's threat landscape, security visibility, incident response, and resource protection are essential for any organization operating in the cloud. This lab series focused on implementing and configuring essential security and monitoring solutions within Microsoft Azure to enhance an organization's cybersecurity posture. As a security engineer in a simulated enterprise environment, I completed Labs [08 to 11](#) which involved deploying and configuring Azure Monitor, enabling Microsoft Defender for Cloud features, applying Just-In-Time (JIT) access controls for virtual machines, and implementing Microsoft Sentinel for threat detection and automation.

The labs provided practical, hands-on experience in using Azure-native security and monitoring tools to improve visibility into infrastructure activity, collect and analyze logs, reduce attack surfaces, and automate threat detection and response. These exercises simulated real-world responsibilities of a cloud security engineer and were designed to reinforce technical skills related to threat prevention, detection, and response in Azure environments.

Lab 08: Azure Monitor – Log Analytics, Storage Account, and Data Collection Rule

Lab Scenario:

As an Azure Security Engineer for a financial technology company, you're responsible for enhancing visibility into Azure virtual machines (VMs) that handle financial transactions and sensitive data. The Chief Information Security Officer (CISO) has tasked you with implementing centralized monitoring to capture security events and performance metrics. To achieve this, you'll deploy Azure Monitor Agent (AMA), configure Log Analytics Workspaces, create a storage account, and define Data Collection Rules (DCRs) for centralized data gathering.

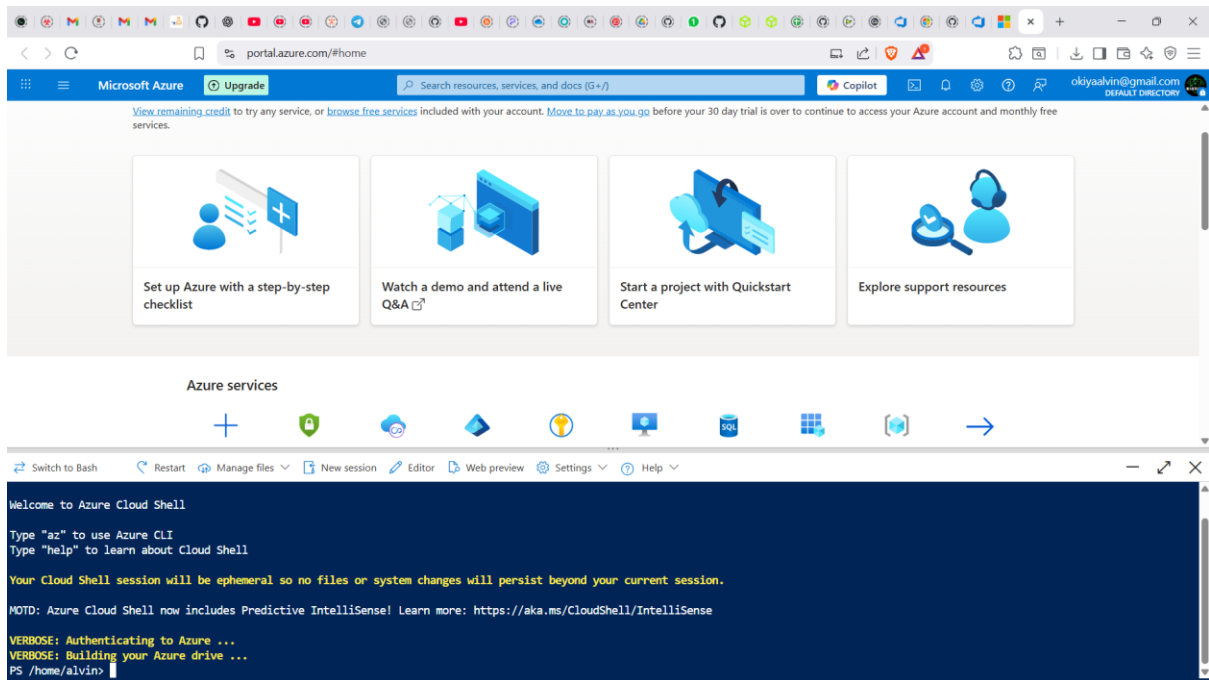
Task Overview:

- **Deploy an Azure Virtual Machine (VM)**
- **Create a Log Analytics Workspace** for log collection
- **Set up an Azure Storage Account** to store diagnostics

- **Create a Data Collection Rule (DCR)** to define what logs to collect and where to send them

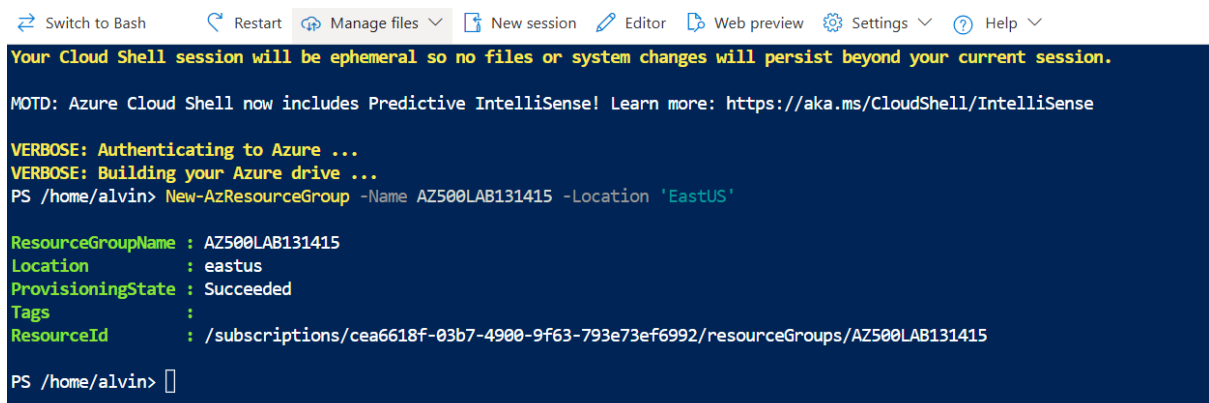
Exercise 1: Deploy an Azure virtual machine

To be able to do this exercise, I started by signing into my azure account via the portal, then opened the cloud shell and ensured that PowerShell is selected:



In the PowerShell session within the Cloud Shell pane, I ran the following command to create a resource group that will be used in this lab:

`New-AzResourceGroup -Name AZ500LAB131415 -Location 'EastUS'`



I then ran the following command to enable encryption at host (EAH)

Register-AzProviderFeature -FeatureName "EncryptionAtHost" -ProviderNamespace Microsoft.Compute

```
Switch to Bash Restart Manage files New session Editor Web preview Settings Help
PS /home/alvin> New-AzResourceGroup -Name AZ500LAB131415 -Location 'EastUS'

ResourceGroupName : AZ500LAB131415
Location           : eastus
ProvisioningState   : Succeeded
Tags               :
ResourceId          : /subscriptions/cea6618f-03b7-4900-9f63-793e73ef6992/resourceGroups/AZ500LAB131415

PS /home/alvin> Register-AzProviderFeature -FeatureName "EncryptionAtHost" -ProviderNamespace Microsoft.Compute

FeatureName      ProviderName      RegistrationState
-----
EncryptionAtHost Microsoft.Compute Registered

PS /home/alvin> 
```

Once that was done, I ran the following to create a new Azure virtual machine:

New-AzVm -ResourceGroupName "AZ500LAB131415" -Name "myVM" -Location 'EastUS' -VirtualNetworkName "myVnet" -SubnetName "mySubnet" -SecurityGroupName "myNetworkSecurityGroup" -PublicIpAddressName "myPublicIpAddress" -PublicIpSku Standard -OpenPorts 80,3389 -Size Standard_D2s_v3

```
portal.azure.com/#home
Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot okiyaalvin@gmail.com
Switch to Bash Restart Manage files New session Editor Web preview Settings Help
PS /home/alvin> New-AzVm -ResourceGroupName "AZ500LAB131415" -Name "myVM" -Location 'EastUS' -VirtualNetworkName "myVnet" -SubnetName "mySubnet" -SecurityGroupName "myNetworkSecurityGroup" -PublicIpAddressName "myPublicIpAddress" -PublicIpSku Standard -OpenPorts 80,3389 -Size Standard_D2s_v3

cmdlet New-AzVm at command pipeline position 1
Supply values for the following parameters:
Credential
User: localadmin
Password for user localadmin: *****

WARNING: Upcoming breaking changes in the cmdlet 'New-AzVm':
The default VM size will change from 'Standard_D2s_v3' to 'Standard_D2s_v5'.
- This change will take effect on '11/1/2025'
- The change is expected to take effect in Az version: '15.0.0'
- The change is expected to take effect in Az.Compute version: '11.0.0'
Note: Go to https://aka.ms/azps-changewarnings for steps to suppress this breaking change warning, and other information on breaking changes in Azure PowerShell.

ResourceGroupName : AZ500LAB131415
Id                : /subscriptions/cea6618f-03b7-4900-9f63-793e73ef6992/resourceGroups/AZ500LAB131415/providers/Microsoft.Compute/virtualMachines/myVM
VmId              : db5b012-0417-4fad-a6b1-b6ac4c2b6da4
Name              : myVM
Type              : Microsoft.Compute/virtualMachines
Location          : eastus
Tags              : {}
HardwareProfile   : {VmSize}
NetworkProfile    : {NetworkInterfaces}
SecurityProfile   : {UefiSettings, SecurityType}
OSProfile         : {ComputerName, AdminUsername, WindowsConfiguration, Secrets, AllowExtensionOperations, RequireGuestProvisionSignal}
ProvisioningState : Succeeded
StorageProfile    : {ImageReference, OsDisk, DataDisks, DiskControllerType, AlignRegionalDisksToVMZone}
FullyQualifiedDomainName : myvm-b59cef.EastUS.cloudapp.azure.com
TimeCreated       : 7/28/2025 8:01:13 AM
Etag              : "2"
```

Once prompted for username I used local admin as username and password I created one.

I then ran the following to confirm that the virtual machine named **myVM** was created and its **ProvisioningState** is **Succeeded**:

Get-AzVM -Name 'myVM' -ResourceGroupName 'AZ500LAB131415' | Format-Table

```
portal.azure.com/#home

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Capilot okiyaalvin@gmail.com
Switch to Bash Restart Manage files New session Editor Web preview Settings Help

The default VM size will change from 'Standard_D2s_v3' to 'Standard_D2s_v5'.
- This change will take effect on '11/1/2025'
- The change is expected to take effect in Az version : '15.0.0'
- The change is expected to take effect in Az.Compute version : '11.0.0'
Note : Go to https://aka.ms/azps-changewarnings for steps to suppress this breaking change warning, and other information on breaking changes in Azure PowerShell.

ResourceGroupName : AZ500LAB131415
Id                : /subscriptions/cea6618f-83b7-4908-9f63-793e73ef6992/resourceGroups/AZ500LAB131415/providers/Microsoft.Compute/virtualMachines/myVM
VmId              : dbd50812-0417-4fad-a6b1-b6ac4c2b6da4
Name              : myVM
Type              : Microsoft.Compute/virtualMachines
Location          : eastus
Tags              : {}
HardwareProfile   : {VmSize}
NetworkProfile    : {NetworkInterfaces}
SecurityProfile   : {UefiSettings, SecurityType}
OSProfile         : {ComputerName, AdminUsername, WindowsConfiguration, Secrets, AllowExtensionOperations, RequireGuestProvisionSignal}
ProvisioningState : Succeeded
StorageProfile    : {ImageReference, OsDisk, DataDisks, DiskControllerType, AlignRegionalDisksToVMZone}
FullyQualifiedDomainName : myvm-b59cef.EastUS.cloudapp.azure.com
TimeCreated       : 7/28/2025 8:01:13 AM
Etag              : "2"

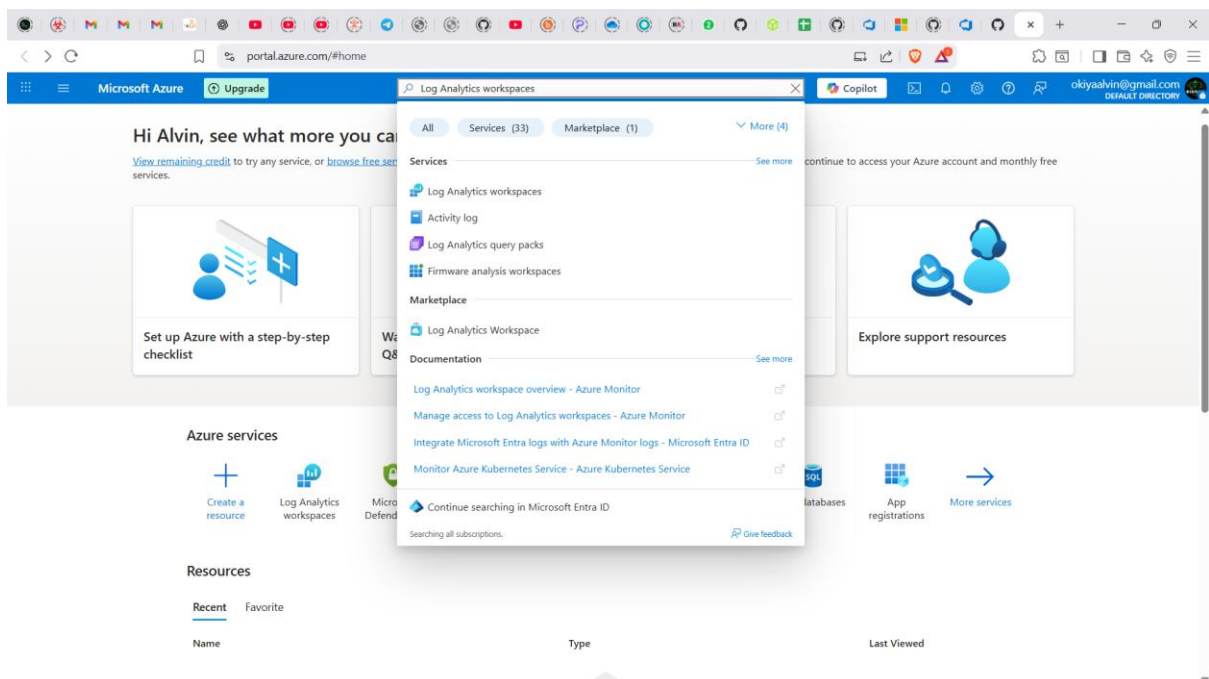
PS /home/alvin> Get-AzVM -Name 'myVM' -ResourceGroupName 'AZ500LAB131415' | Format-Table

ResourceGroupName Name Location VmSize OsType NIC ProvisioningState
-----
AZ500LAB131415 myVM eastus Standard_D2s_v3 Windows myVM Succeeded

PS /home/alvin>
```

Exercise 2: Create a Log Analytics workspace

To create a log analytics workspace, I started by navigating to the search bar on my azure portal and searched for log analytics workspace as shown below, then selected it.



On opening the interface I clicked create, which redirected me to the following interface:

portal.azure.com/#create/Microsoft.LogAnalyticsOMS

Microsoft Azure Upgrade Search resources, services, and docs (G+I)

Home > Log Analytics workspaces >

Create Log Analytics workspace

Basics Tags Review + Create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * AZ500LAB131415 [Create new](#)

Instance details

Name * Lab8-Log-Analytics-workspace

Region * East US

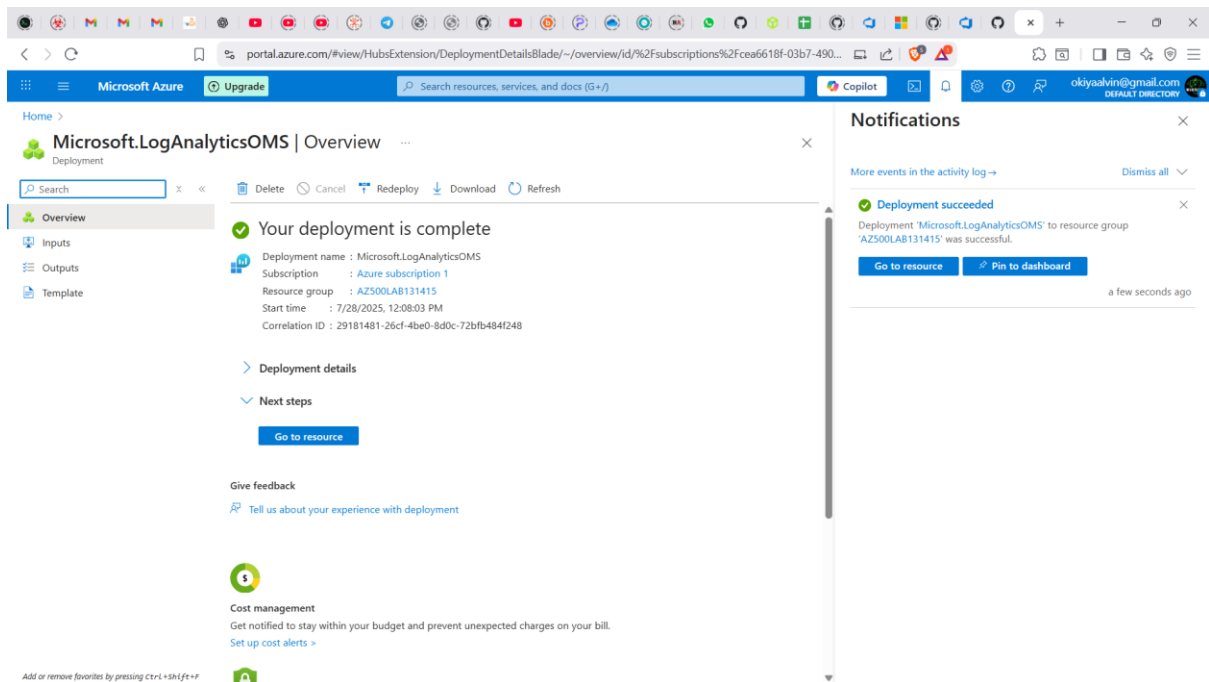
[Review + Create](#) [Previous](#) [Next: Tags >](#)

On this interface I filled the following details:

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Resource group	AZ500LAB131415
Name	any valid, globally unique name
Region	East US

I then clicked review and create, then create

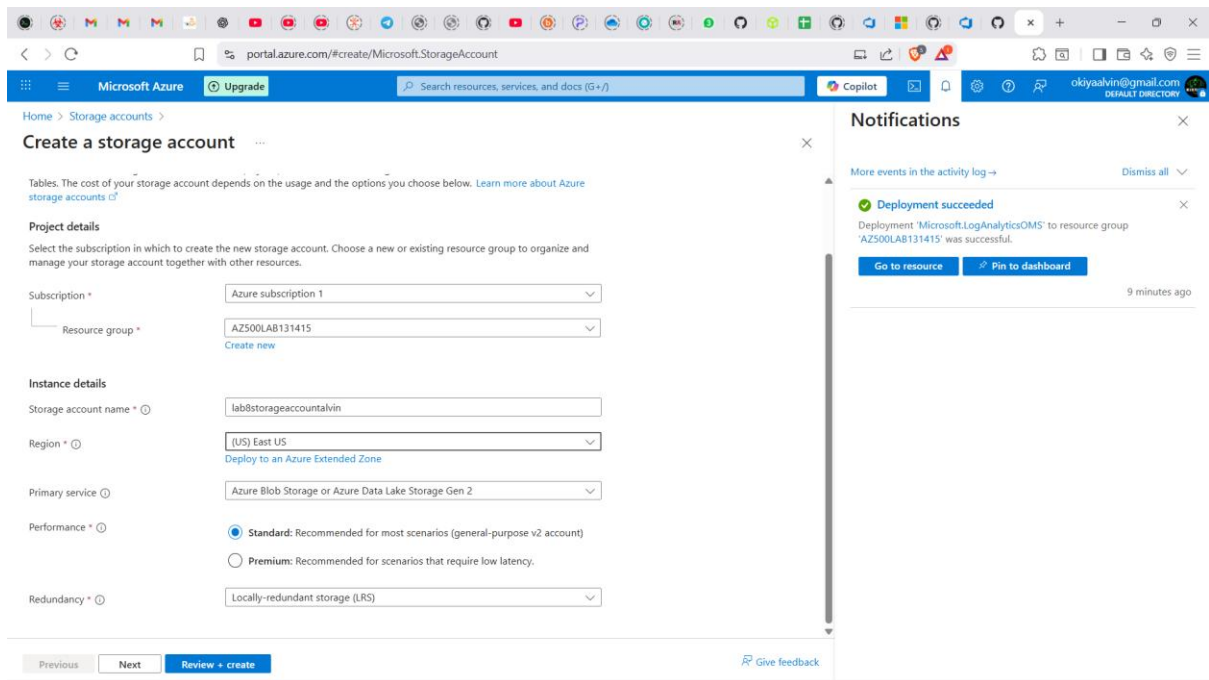
The deployment initialized and after some few minutes it was complete:



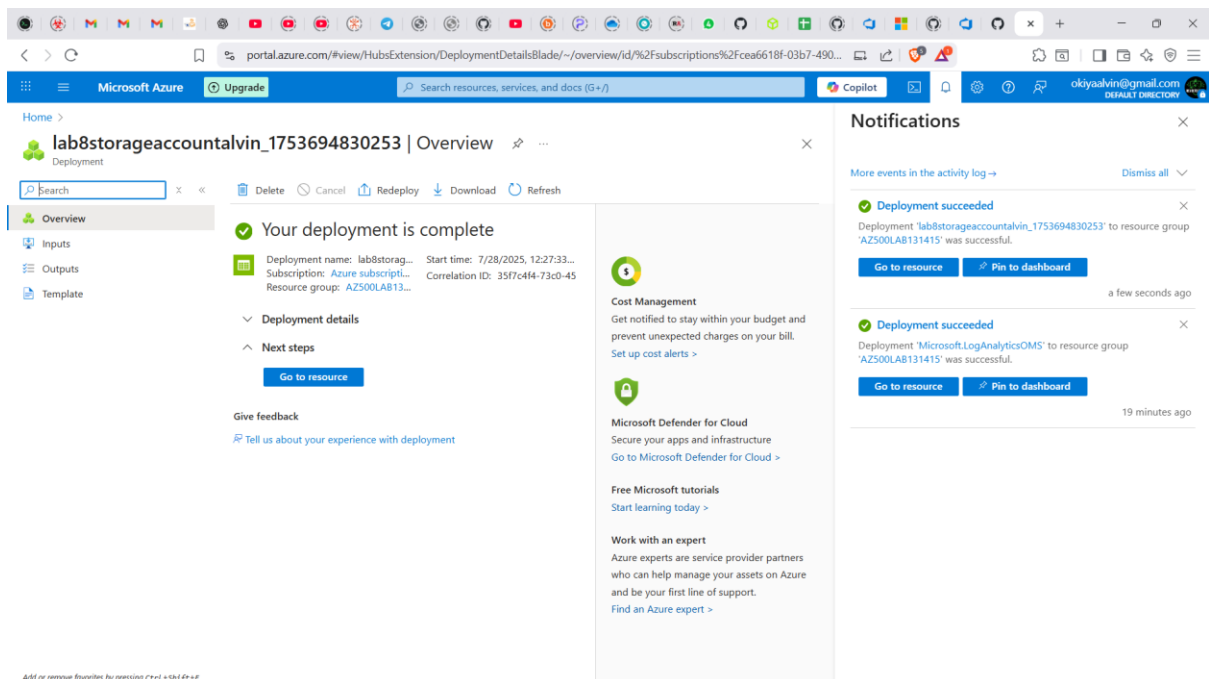
Exercise 3: Create an Azure storage account

To create a storage account, I started by navigating to the search bar on my azure portal and Searching for Storage account, and on the storage account blade, I clicked the create button and filled the basics tab with the following details:

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Resource group	AZ500LAB131415
Instance details	
Storage account name	any globally unique name between 3 and 24 in length consisting of letters and digits
Region	(US) EastUS
Primary service	Azure Blob Storage or Azure Data Lake Storage Gen 2
Performance	Standard (general-purpose v2 account)
Redundancy	Locally redundant storage (LRS)

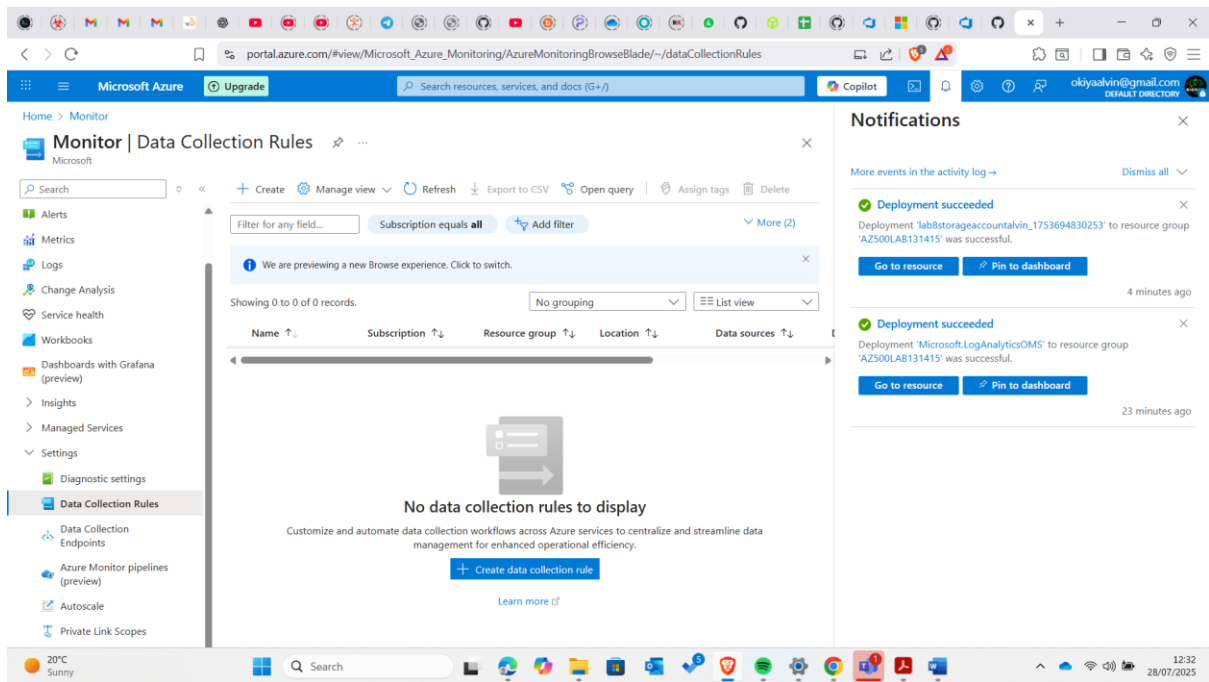


On the **Basics** tab of the **Create storage account** blade, I clicked **Review + create**. After the validation process completes, I clicked **Create**. The deployment started and the storage account was created.



Exercise 4: Create a Data Collection Rule

To perform this task, I navigated to the search bar on my azure portal and searched **Monitor** then selected Monitor on the dropdown. On the monitor settings blade, I clicked on **Data collection rules> + create**:



On the basics tab of create data collection rule I specified the following settings:

Setting	Value
Rule details	
Rule Name	DCR1
Subscription	the name of the Azure subscription you are using in this lab
Resource Group	AZ500LAB131415
Region	East US
Platform Type	Windows
Data Collection Endpoint	Leave Blank

Then clicked on “Next: Resources”:

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/I)

Home > Monitor | Data Collection Rules >

Create Data Collection Rule

Data collection rule management

To create a Data Collection Rule that collects platform metrics, click here.

Basics | Resources | Collect and deliver | Tags | Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources. [Learn more](#)

Rule details

Rule Name *

Subscription *

Resource Group * [Create new](#)

Region *

Platform Type * ☒ Windows ☐ Linux ☐ All

Data Collection Endpoint

[Review + create](#) [< Previous](#) [Next: Resources >](#)

Notifications

More events in the activity log → [Dismiss all](#)

Deployment succeeded

Deployment 'lab8storageaccountalvin_1753694830253' to resource group 'AZ500LAB131415' was successful.

[Go to resource](#) [Pin to dashboard](#)

8 minutes ago

Deployment succeeded

Deployment 'Microsoft.LogAnalyticsOMS' to resource group 'AZ500LAB131415' was successful.

[Go to resource](#) [Pin to dashboard](#)

28 minutes ago

Then clicked on “+ Add Resources”, on the scope, I checked the subscription am using currently, then clicked on apply

Microsoft Azure | Upgrade | Search resources, services, and docs (G+/I)

Home > Monitor | Data Collection Rules >

Create Data Collection Rule

Data collection rule management

To create a Data Collection Rule that collects platform metrics, click here.

Basics | **Resources** | Collect and deliver | Tags | Review + create

Pick a set of resources to collect data from. The Azure Monitor Agent will be automatically installed on virt For Windows 10 and 11 devices, [download the client installer](#) and follow the [guidance](#)

This will also enable System Assigned Managed Identity on these resources, in addition to existing User Assigned Id

+ Add resources + Create endpoint

Enable Data Collection Endpoints ☐

Only resources in the same region can be assigned to the same endpoint. [Learn more](#)

Name	Type	Location
No resources found.		

Select a scope

Browse Recent

Resource group Resource types Locations

Search to filter items...

Scope	Resource type	Location
☒ Azure subscription 1	Subscription	-
☐ AZ500LAB131415	Resource group	-

[Apply](#) [Cancel](#) [Clear all selections](#)

[Review + create](#) [< Previous](#) [Next: Collect and deliver >](#)

At the bottom of the **Resources** page, I selected **Next: Collect and deliver >**

Then clicked **+Add Data source** then on the **Add data source** page, changed the **Data source type** drop-down menu to display **Performance Counters**. Left the following default settings:

Setting	Value
Performance counter	Sample rate (seconds)
CPU	60
Memory	60
Disk	60
Network	60

portal.azure.com/#view/Microsoft_Azure_Monitoring/CreateDataCollectionRulesViewModel/_provisioningContext~/~/67B*...

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Monitor | Data Collection Rules >

Create Data Collection Rule

Data collection rule management

To create a Data Collection Rule that collects platform metrics, click here.

Basics Resources **Collect and deliver** Tags Review + create

Configure which data sources to collect and where to send the data to.

+ Add data source

Data source	Destination(s)
No standard data sources or destinations found.	

This data collection rule doesn't have any data sources or destinations selected.

Add data source

* Data source Destination

Select which data source type and the data to collect for your resource(s).

Data source type *

Performance Counters

Choose Basic to enable the collection of performance counters. Choose Custom if you want more control over which performance counters are collected.

None Basic Custom

Performance counter	Sample rate (seconds)
☒ CPU	60
☒ Memory	60
☒ Disk	60
☒ Network	60

Review + create < Previous Next : Tags >

Add data source Next : Destination > Cancel

I then clicked on the button labelled **Next: Destination >** to proceed

I then Clicked **+ Add destination**, changed the **Destination type** drop-down menu to display **Azure Monitor Logs**. In the **Subscription** window, I ensured that my *Subscription* is displayed, then changed the **Account or namespace** drop-down menu to reflect the previously created Log Analytics Workspace.

I then clicked on **Add data source** at the bottom of the page:

portal.azure.com/#view/Microsoft_Azure_Monitoring/CreateDataCollectionRulesViewModel/_provisioningContext~/7B*...

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Monitor | Data Collection Rules >

Create Data Collection Rule

Data collection rule management

To create a Data Collection Rule that collects platform metrics, click here.

Basics Resources **Collect and deliver** Tags Review + create

Configure which data sources to collect and where to send the data to.

+ Add data source

Data source	Destination(s)
No standard data sources or destinations found.	

This data collection rule doesn't have any data sources or destinations selected.

Review + create < Previous Next: Tags >

Add data source

* Data source Destination

Select the destination(s) for where the data will be delivered. Normal usage charges for the destination will occur. [Learn more about pricing.](#)

+ Add destination

* Destination type	Subscription	Destination Details
Azure Monitor Logs	Azure subscription 1	Lab8-Log-Analytics-workspace (...)

Add data source < Previous Cancel

Then review and create, then create:

portal.azure.com/#view/Microsoft_Azure_Monitoring/CreateDataCollectionRulesViewModel/_provisioningContext~/7B*...

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Monitor | Data Collection Rules >

Create Data Collection Rule

Data collection rule management

Validation passed

To create a Data Collection Rule that collects platform metrics, click here.

Basics Resources **Collect and deliver** Tags **Review + create**

Basics

Data rule name	DCR1
Subscription	Azure subscription 1
Resource Group	AZ500LAB131415

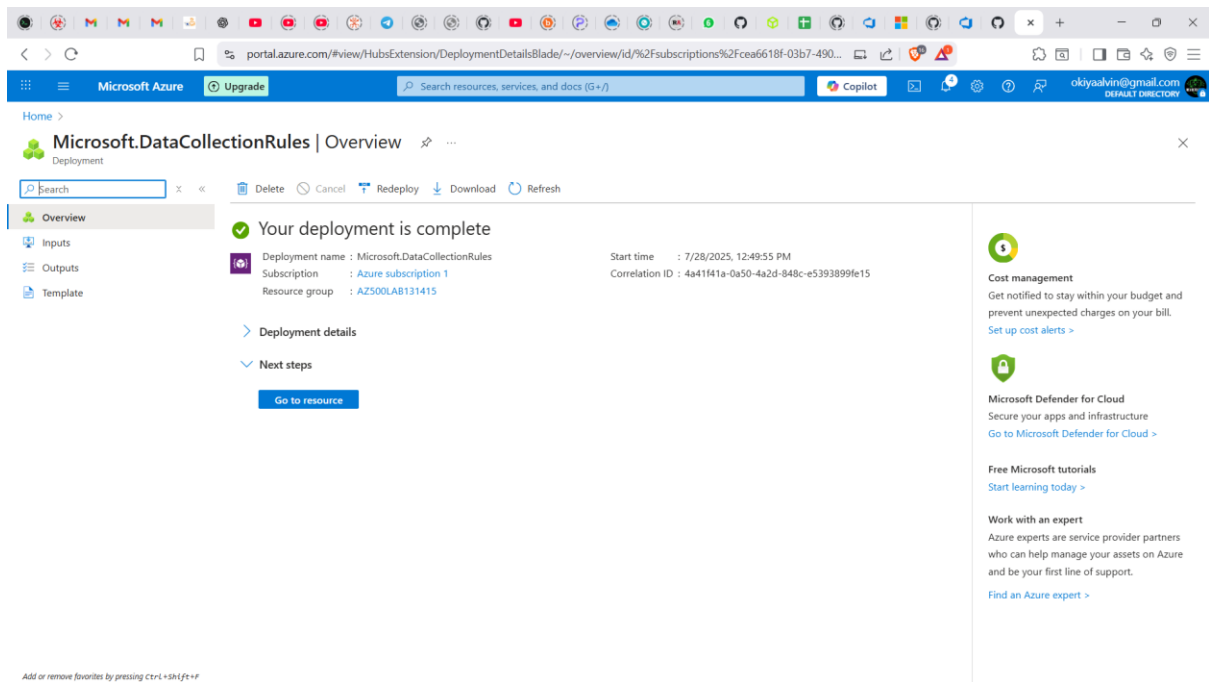
Selected resources

Resources	Type
myvm	microsoft.compute/virtualmachines

Showing 1 - 1 of 1 results.

Create < Previous Next: >

This initialised the deployment process:



Having done this, I have successfully deployed an Azure virtual machine, Log Analytics workspace, Azure storage account, and a data collection rule to collect events and performance counters from virtual machines with Azure Monitor Agent.

I did not clean up the resources from this lab, as they are needed for the Microsoft Defender for Cloud lab, the 'Enable just-in-time access on VMs' lab, and the Microsoft Sentinel lab, that is lab 9, 10 and 11.

Lab 09: Microsoft Defender for Cloud – Enhanced Security for Servers

Lab Scenario:

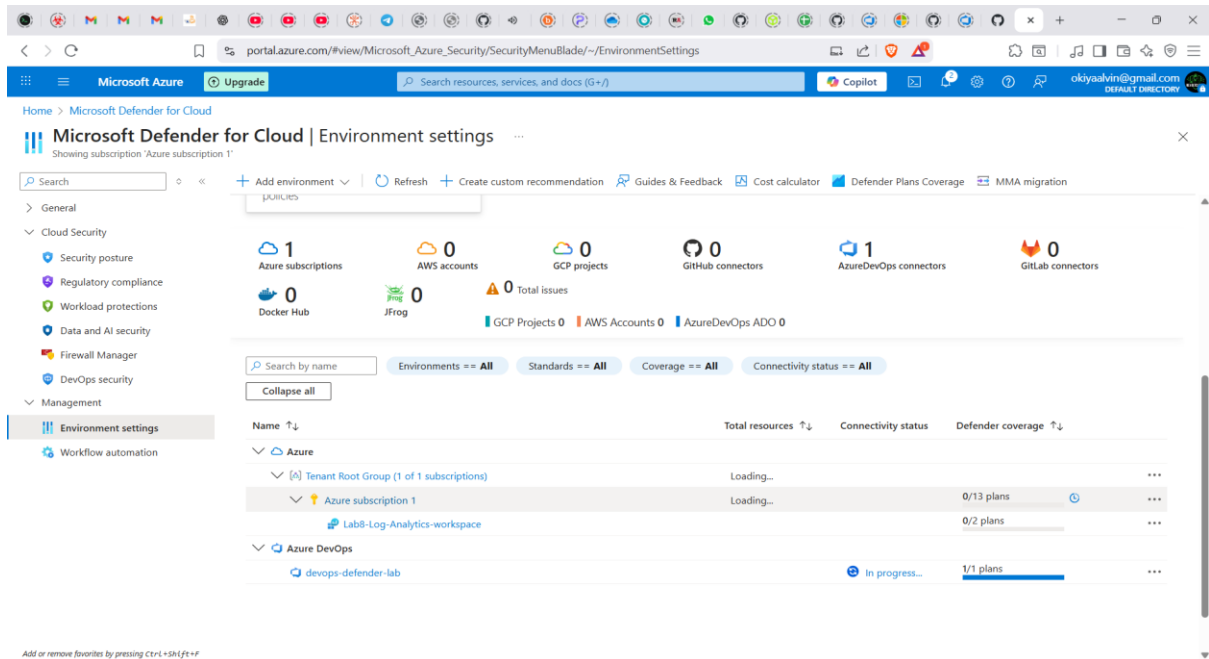
Working as a Security Engineer for a global e-commerce company, you're responsible for fortifying Azure VMs and hybrid servers against threats. Following concerns raised by the CISO, you're instructed to enable enhanced security features using Microsoft Defender for Cloud. This includes activating Defender for Servers Plan 2, which delivers advanced threat detection, security monitoring, and vulnerability management.

Task Overview:

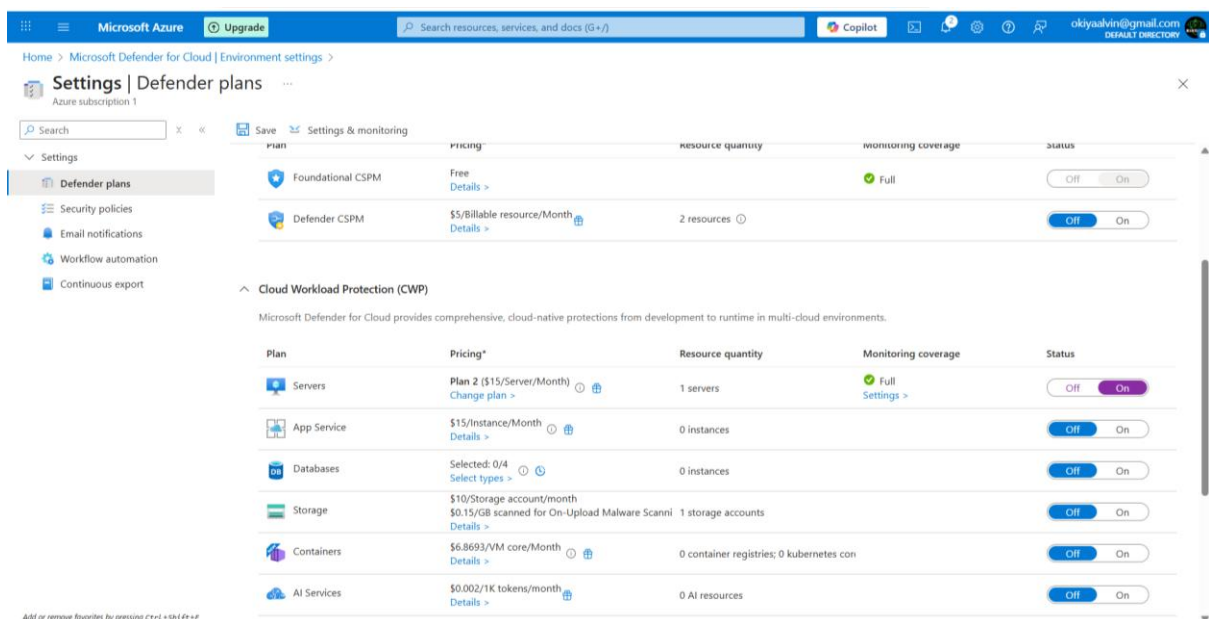
- **Access Microsoft Defender for Cloud**
- **Enable Microsoft Defender for Servers Plan 2**
- **Review security capabilities and features of the enhanced protection plan**

Configure Microsoft Defender for Cloud Enhanced Security Features for Servers

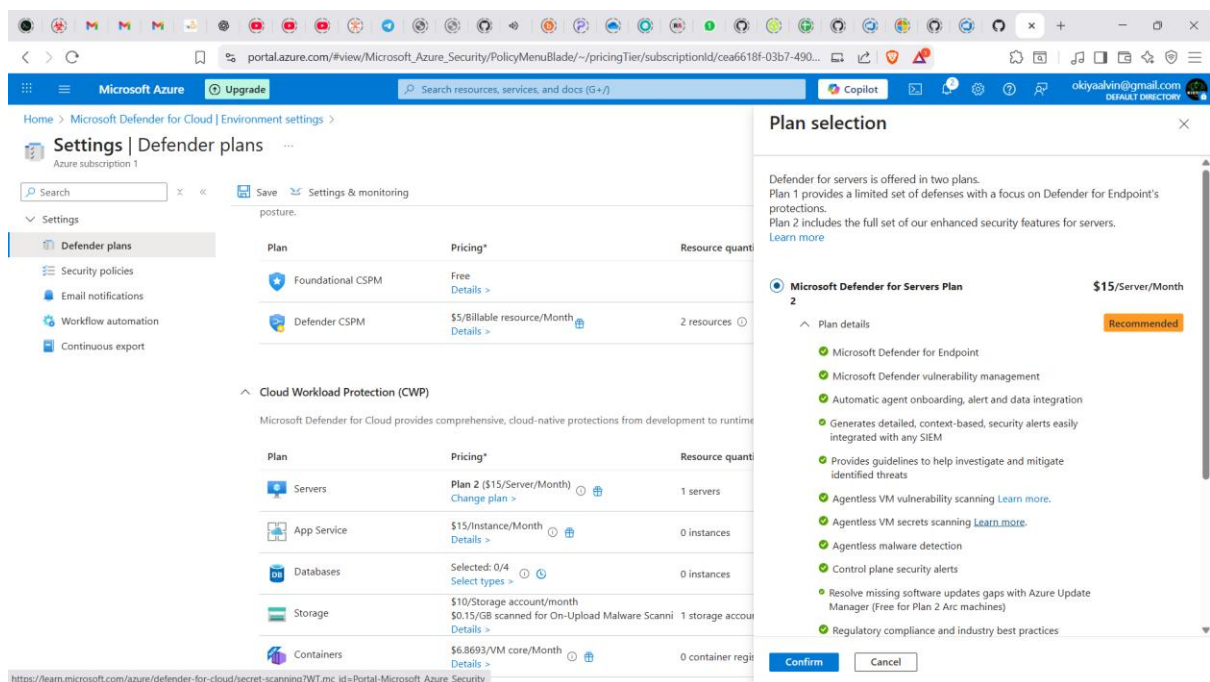
I started by navigating to the search bar on my azure portal and typing **Microsoft Defender for Cloud** then selecting it from the search results. On the **Microsoft Defender for Cloud, Management blade**, I navigated to the **Environment settings** and expanded the environment settings folders until the **subscription** section is displayed, then I clicked the **subscription** to view details:



After clicking on my subscription, it opened a new window for Defender plans, In the **Settings** blade, under **Defender plans**, I expanded **Cloud Workload Protection (CWP)** then selected servers, then on the right side of the page, I changed the **Status** from **Off** to **On**, then clicked **Save**.



To review the details of **Microsoft Defender for Servers Plan 2**, select **Change plan >**.



Note: Enabling the Cloud Workload Protection (CWP) Servers plan from Off to On enables Microsoft Defender for Servers Plan 2.

By following those steps, I have enabled Microsoft Defender for Servers Plan 2 on my subscription.

Lab 10: Enable Just-in-Time (JIT) Access on VMs

Lab Scenario:

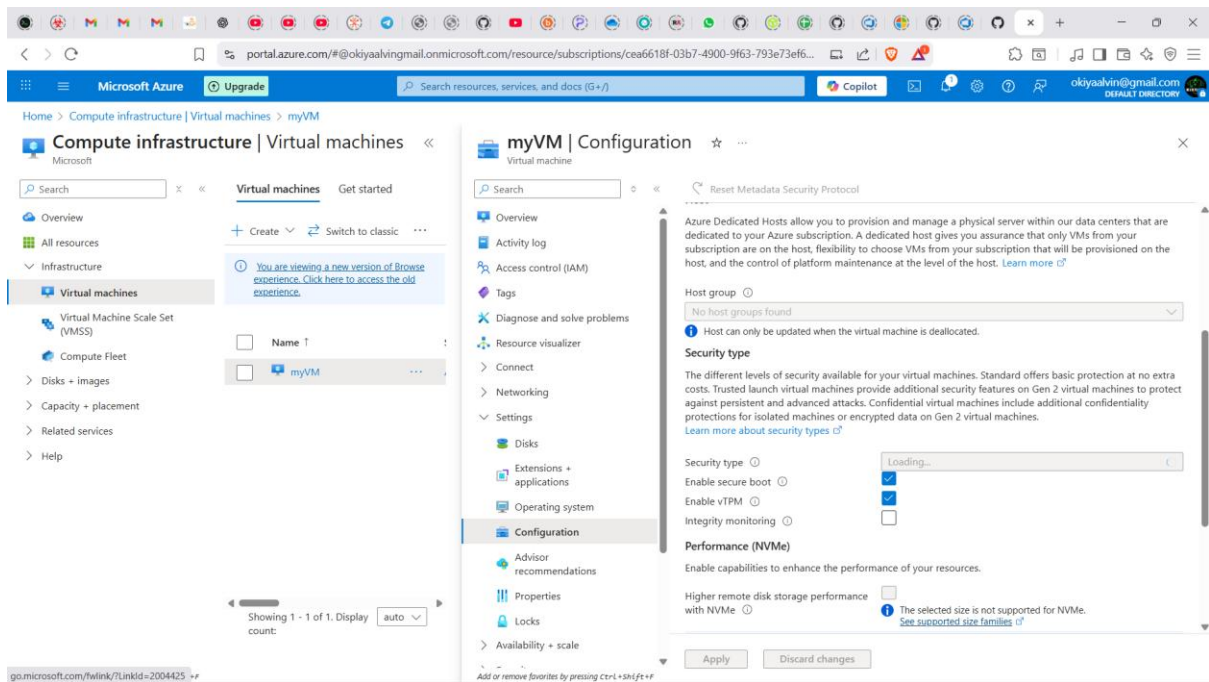
In your role at a financial services company, you are tasked with reducing exposure of Azure VMs used for sensitive applications. Continuous access to these VMs poses risks such as brute-force attacks. You must implement JIT access policies to minimize these threats by granting time-limited, IP-restricted access to VM ports.

Task Overview:

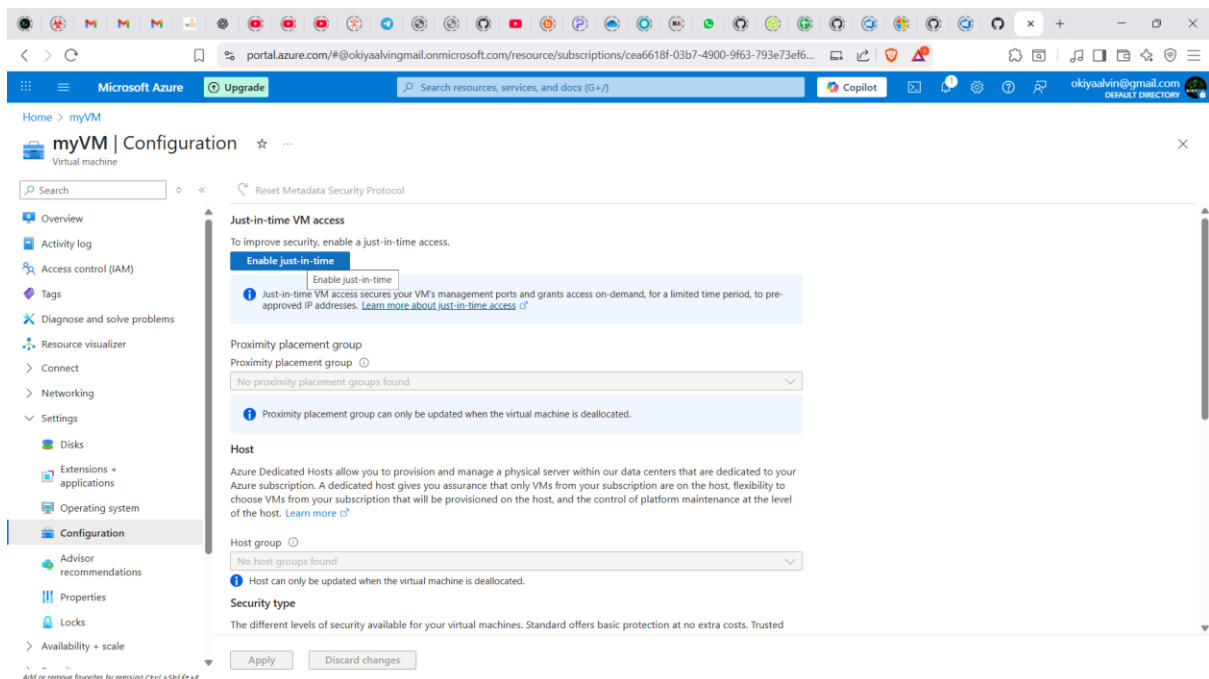
- **Enable Just-in-Time VM Access** via Microsoft Defender for Cloud
- **Customize allowed ports and time limits for JIT access**
- **Request and test access to the JIT-enabled VM**

Exercise 1: Enable JIT on your VMs from Azure virtual machines

To do this exercise I started by navigating to the search bar and searching for Virtual Machines, then selected my VMs. Then Selected **Configuration** from the **Settings** section of **myVM**.



Under **Just-in-time VM access**, I selected **Enable just-in-time**



At this point you might encounter an error in the notification section telling you, you lack sufficient permissions if you are using a free subscription.

 **Error enabling just-in-time access**

You do not have permissions to enable just-in-time access for 'myVM'.
Contact the subscription owner to get permissions.

 **Help me troubleshoot**

2 minutes ago

Troubleshooting

To tackle this, either switch to Pay as you Go, or if you are a student, Request your instructor for cloud slices to be able to work on the assignment.

In my case I used the cloud slices where I repeated all the previous steps before proceeding.

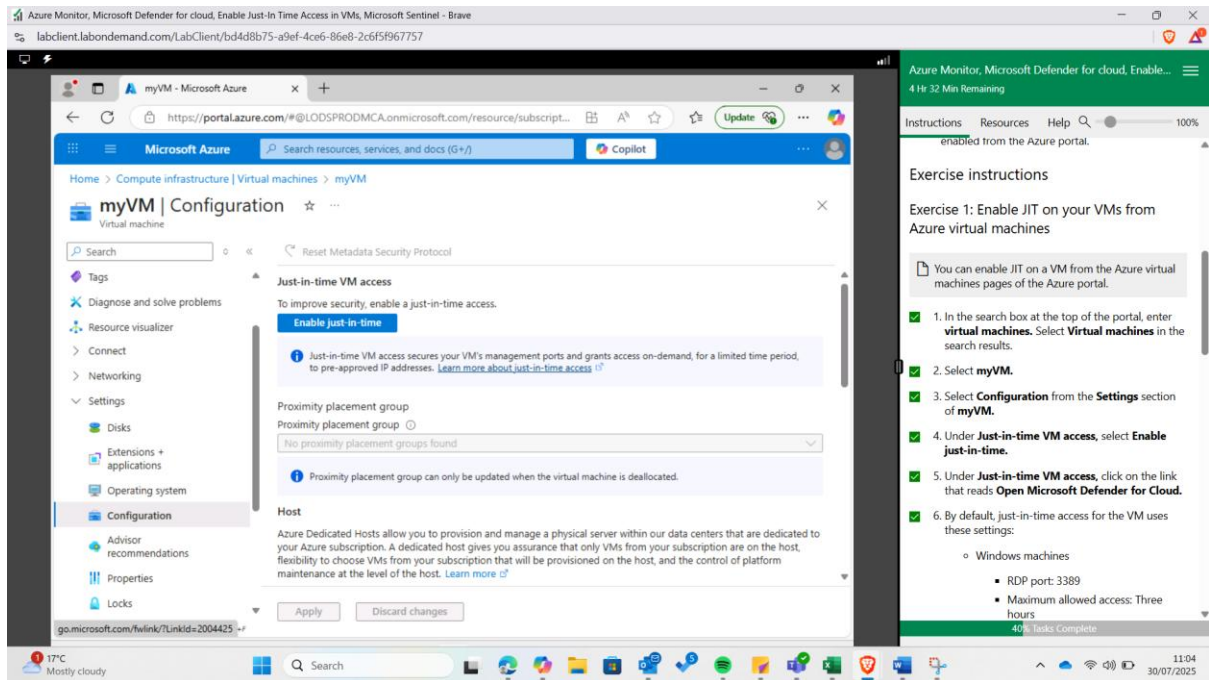
When using cloud slices, you may come across such an error during tackling lab 8, in such a case close the session and relaunch the lab afresh.

The screenshot shows a Windows desktop with a taskbar at the bottom. The main window is the Azure portal, displaying the 'Subscriptions' page. A Cloud Shell session is open, showing a PowerShell command to create a new VM. The command is: `New-AzVM -ResourceGroupName 'AZ500LAB131415' -Name 'myVM' -Location 'EastUS' -VirtualNetworkName 'myVNet' -SubnetName 'mySubnet' -SecurityGroupName 'myNetworkSecurityGroup' -PublicIpAddressName 'myPublicIpAddress' -OpenPorts 80,3389 -Size Standard_D2s_v3`. The output shows an error: `New-AzVM: A parameter cannot be found that matches parameter name 'PublicIpSku'.` The error message is highlighted in red. The Cloud Shell session is titled 'New-AzVM at command pipeline position 1' and shows the command 'New-AzVM -ResourceGroupName 'AZ500LAB131415' -Name 'myVM' -Location 'EastUS' -VirtualNetworkName 'myVNet' -SubnetName 'mySubnet' -SecurityGroupName 'myNetworkSecurityGroup' -PublicIpAddressName 'myPublicIpAddress' -OpenPorts 80,3389 -Size Standard_D2s_v3'.

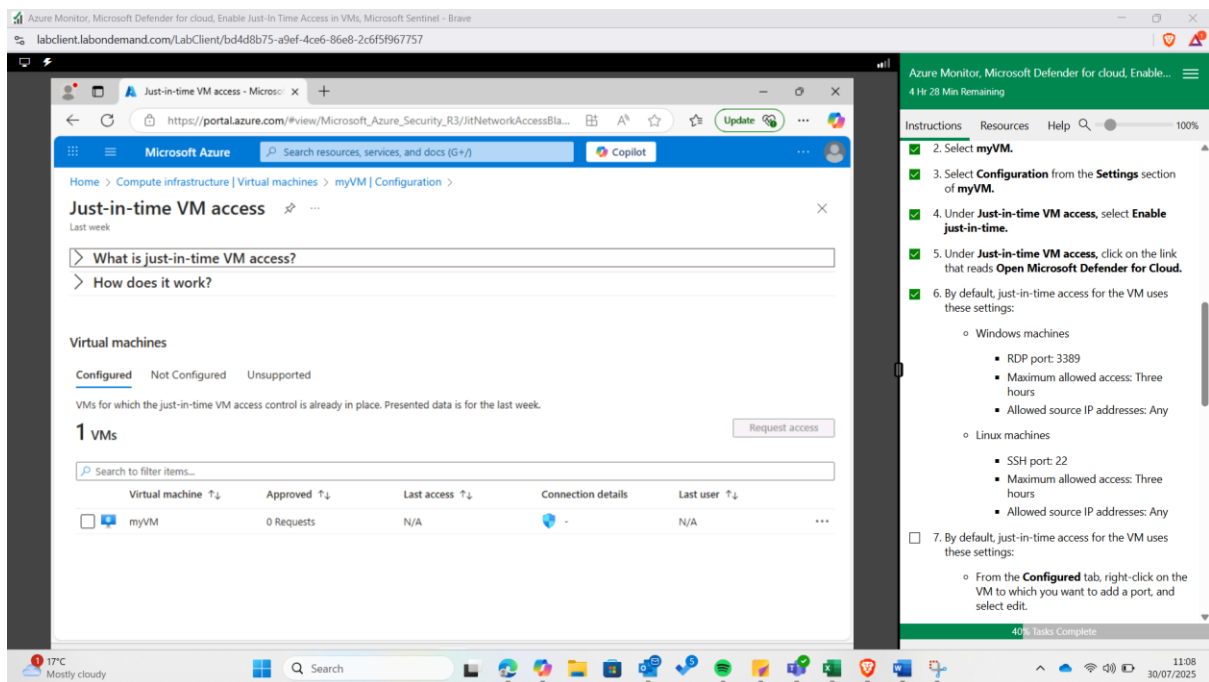
This solves the issue and the command will run successfully as shown:

The screenshot shows a Windows desktop with a taskbar at the bottom. The main window is the Azure portal, displaying the 'Resource groups' page. A Cloud Shell session is open, showing the output of the 'New-AzVM' command. The output is: `ResourceGroupName : AZ500LAB131415
Id : /subscriptions/896bfac5-310d-43fc-958f-0a6cf9fee449/resourceGroups/AZ500LAB131415/providers/Microsoft.Compute/virtualMachines/myVM
Vid : f03fe7f1-be67-4d14-9fb2-399613c712a7
Name : myVM
Type : Microsoft.Compute/virtualMachines
Location : eastus
Tags : {}
HardwareProfile : {VhSize}
NetworkProfile : {NetworkInterfaces}
SecurityProfile : {UefiSettings, SecurityType}
OSProfile : {ComputerName, AdminUsername, WindowsConfiguration, Secrets, AllowExtensionOperations, RequireGuestProvisionSignal}
ProvisioningState : Succeeded
StorageProfile : {ImageReference, OsDisk, DataDisks, DiskControllerType, AlignRegionalDisksToVMZone}
FullyQualifiedDomainName : myvm-f2ecf8.EastUS.cloudapp.azure.com`

On my cloudslice, upon reaching the section that I was previously stuck as I was using my free subscription, I was now able to enable Just In Time (JIT)



After enabling, I then clicked on the link that reads, open Microsoft defender for cloud which open the following interface:

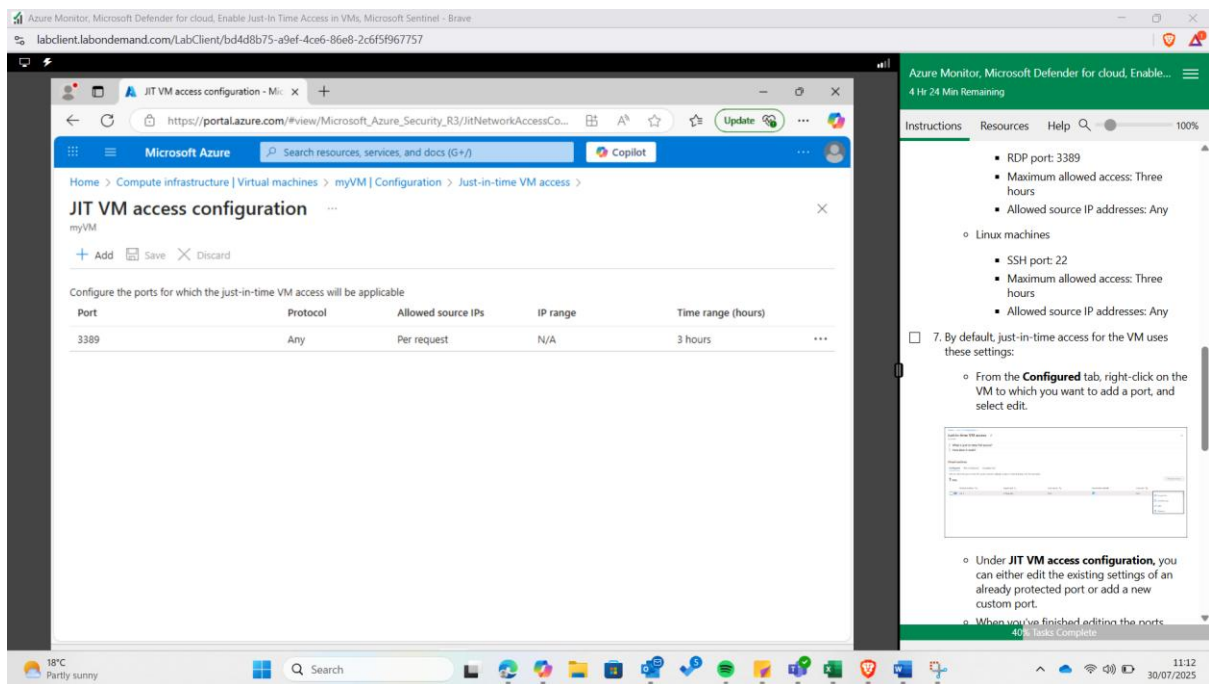


By default, just-in-time access for the VM uses these settings:

- Windows machines
 - RDP port: 3389

- Maximum allowed access: Three hours
- Allowed source IP addresses: Any
- Linux machines
 - SSH port: 22
 - Maximum allowed access: Three hours
 - Allowed source IP addresses: Any

To verify this, I clicked on the 3 dots at the right of the vm, then clicked edit:

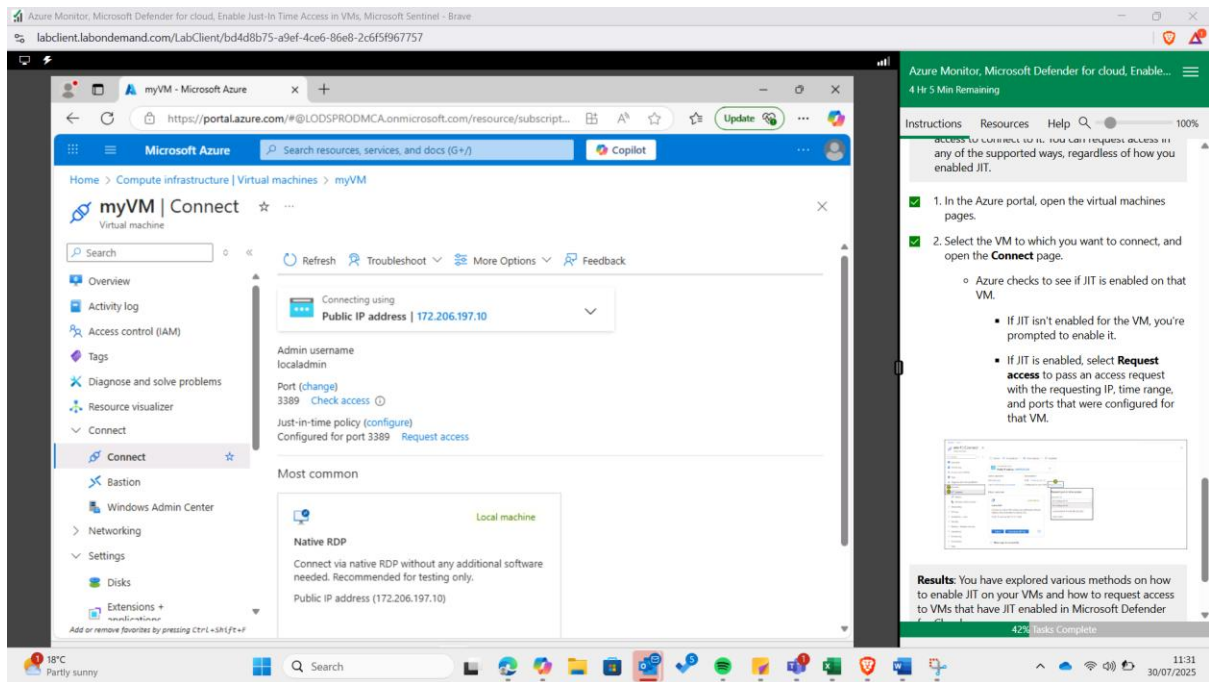


Exercise 2: Request access to a JIT-enabled VM from the Azure virtual machine's connect page.

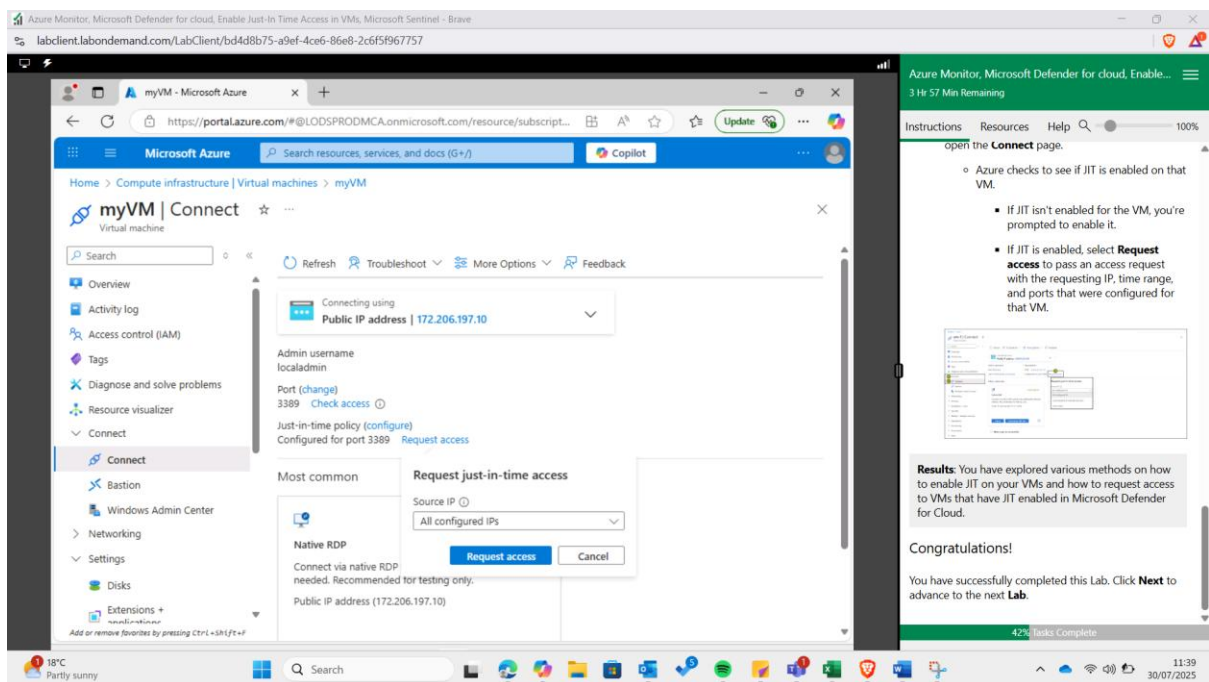
Note: When a VM has a JIT enabled, you have to request access to connect to it. You can request access in any of the supported ways, regardless of how you enabled JIT.

On the azure portal I navigated to the VMs section and selected the VM I want to connect to, which is myVM then opened the connect page:

On opening the connect page, Azure checks to see if JIT is enabled on that VM, if not, you are prompted to enable. If it is enabled you will see an option to request access as shown:



In my case I selected **Request access** to pass an access request with the requesting IP, time range, and ports that were configured for that VM.



Having done this I have explored various methods on how to enable JIT on my VMs and how to request access to VMs that have JIT enabled in Microsoft Defender for Cloud.

Lab 11: Microsoft Sentinel – Threat Detection and Automated Response

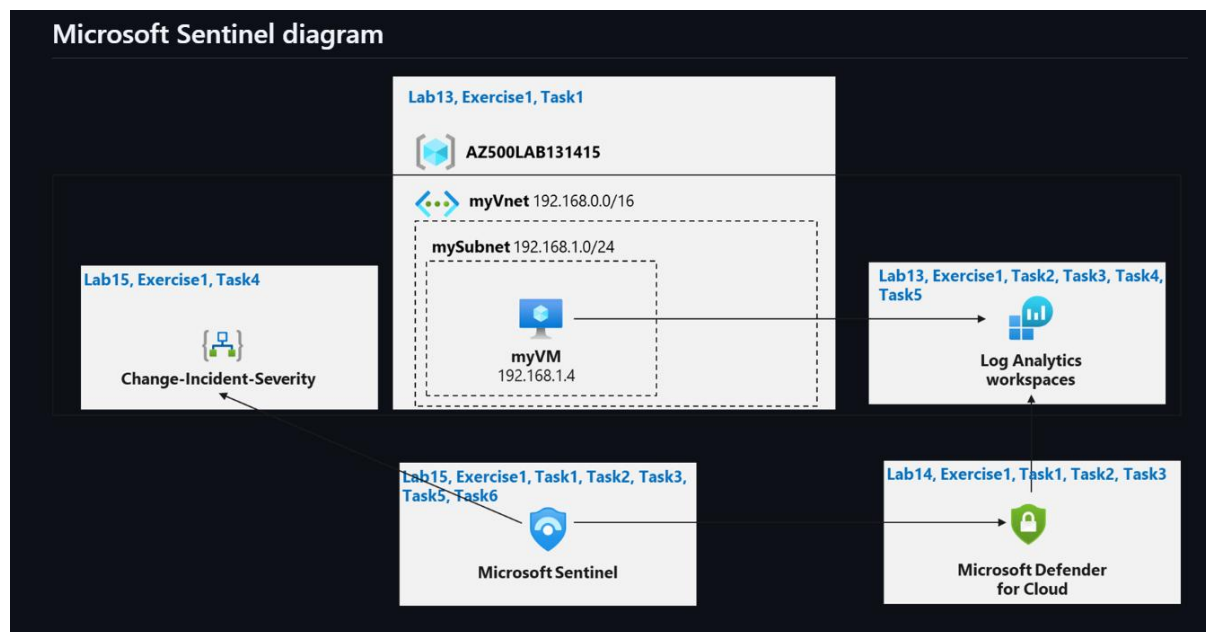
Lab Scenario:

You are required to build a proof-of-concept security monitoring and response system using Microsoft Sentinel. The goal is to collect data from key services, create actionable alerts, and automate incident response using playbooks. This setup will enhance your organization's ability to detect and respond to threats efficiently.

Task Overview:

- **Onboard Microsoft Sentinel and connect it to Log Analytics**
- **Enable Azure Activity and Microsoft Defender for Cloud connectors**
- **Create analytics rules and automated playbooks**
- **Trigger an incident (e.g., deleting a JIT policy) and validate response**

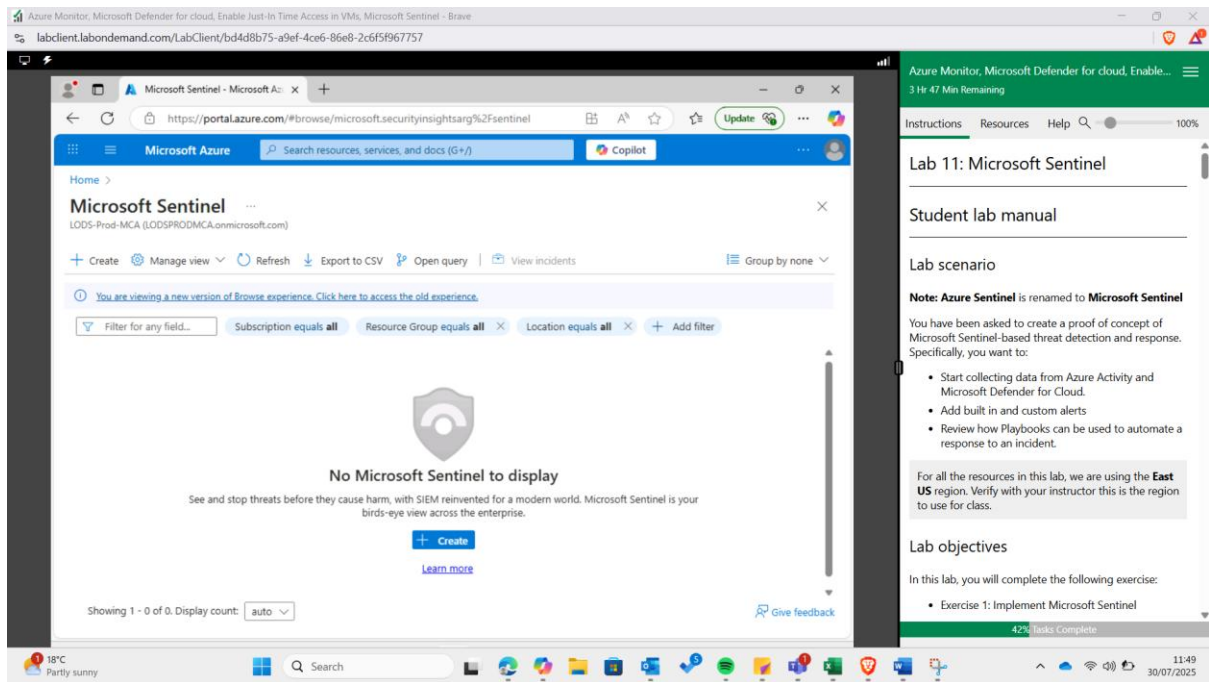
Exercise 1: Implement Microsoft Sentinel



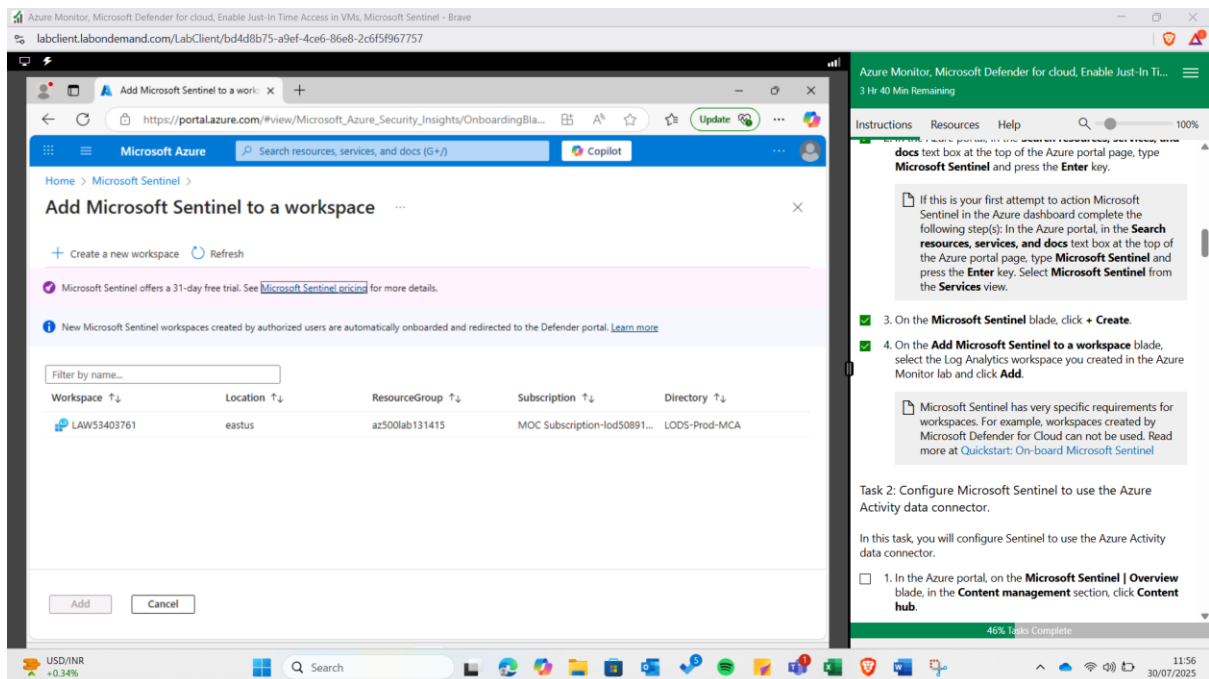
Task 1: On-board Microsoft Sentinel

In this task, I on-boarded Microsoft Sentinel and connected the Log Analytics workspace.

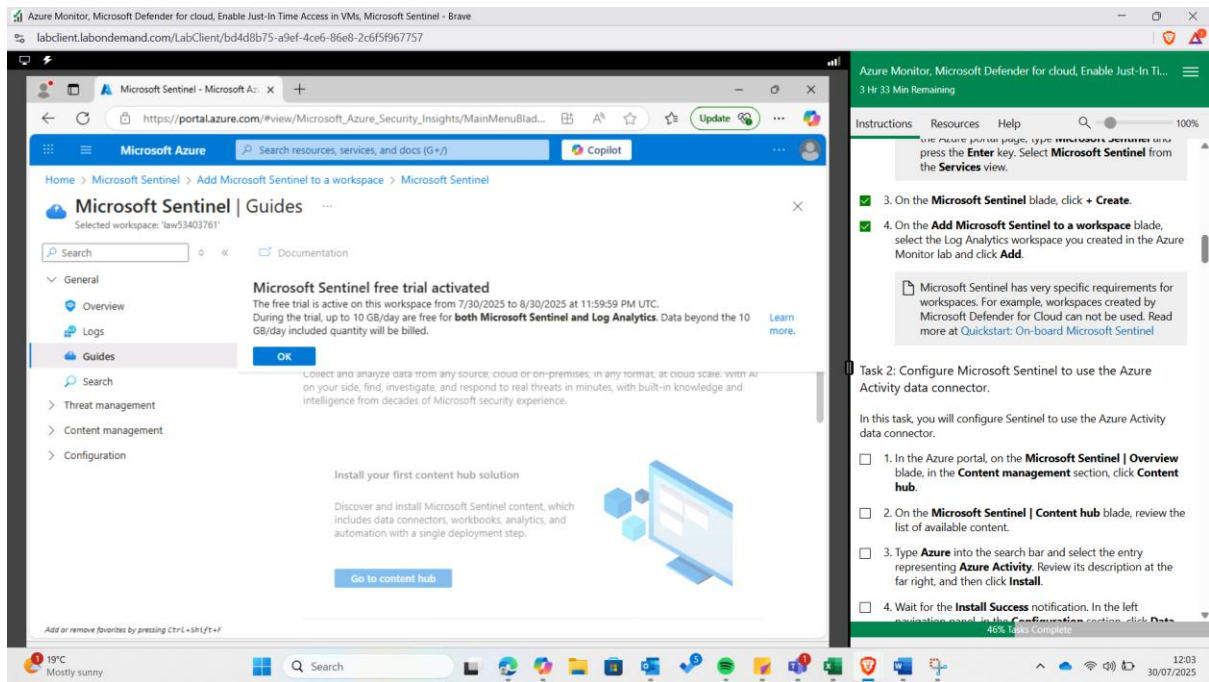
I started by navigating to the search bar in the azure portal and searching for **Microsoft Sentinel**. Then selecting it from the drop down list, and the following page opened up, Where I clicked create:



Upon clicking create, the following interface came up:



On the **Add Microsoft Sentinel to a workspace** blade, I selected the Log Analytics workspace I created in the Azure Monitor lab 08 and clicked **Add**.

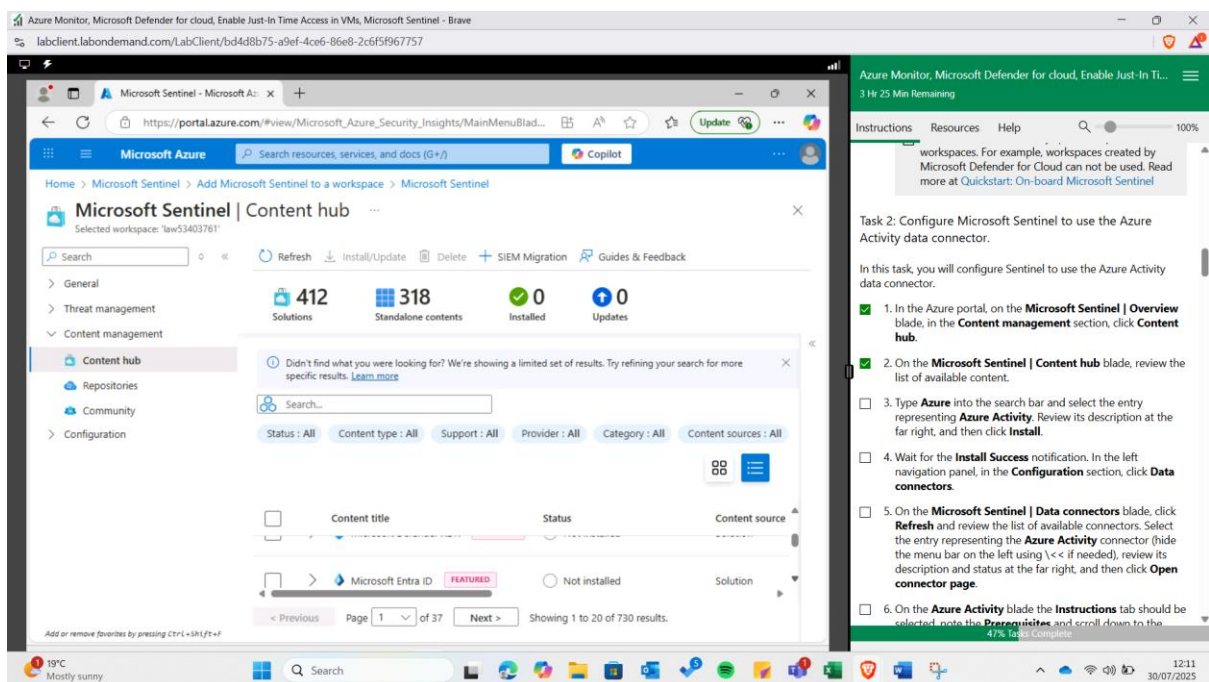


Note: Microsoft Sentinel has very specific requirements for workspaces. For example, workspaces created by Microsoft Defender for Cloud can not be used. Read more at [Quickstart: On-board Microsoft Sentinel](#).

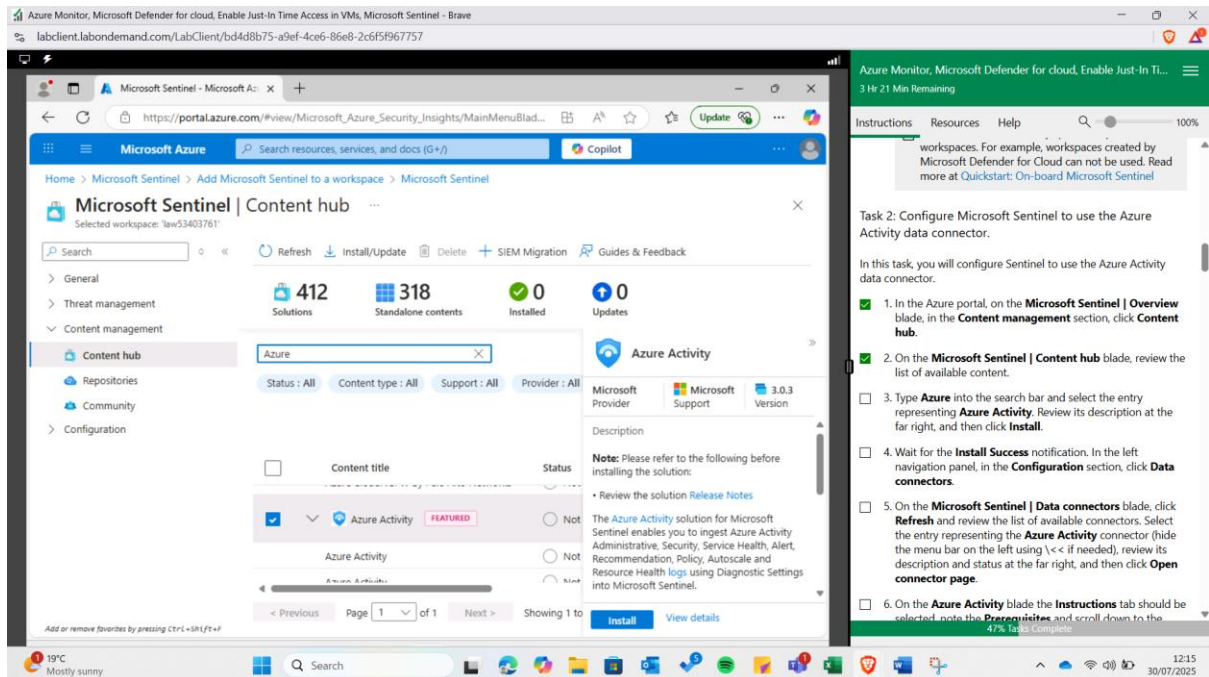
Task 2: Connect Azure Activity to Sentinel

In this task, I configured Sentinel to use the Azure Activity data connector.

In the Azure portal, on the **Microsoft Sentinel | Overview** blade, in the **Content management** section, I clicked **Content hub** then reviewed the list of available content.

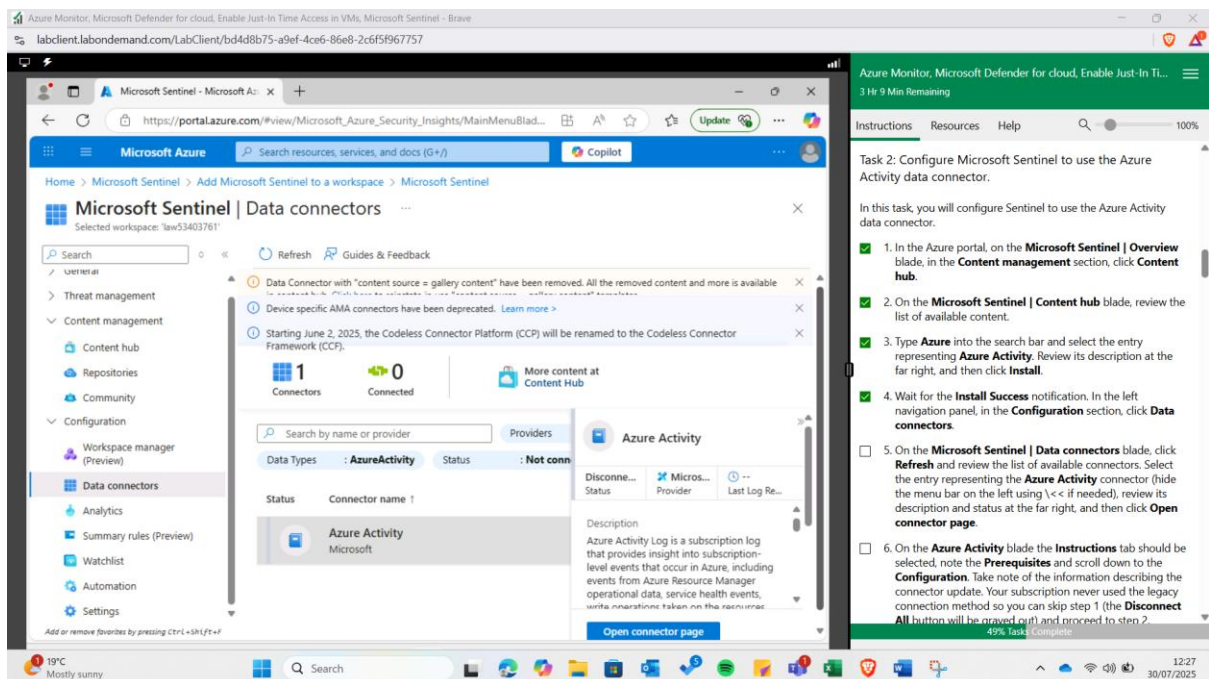


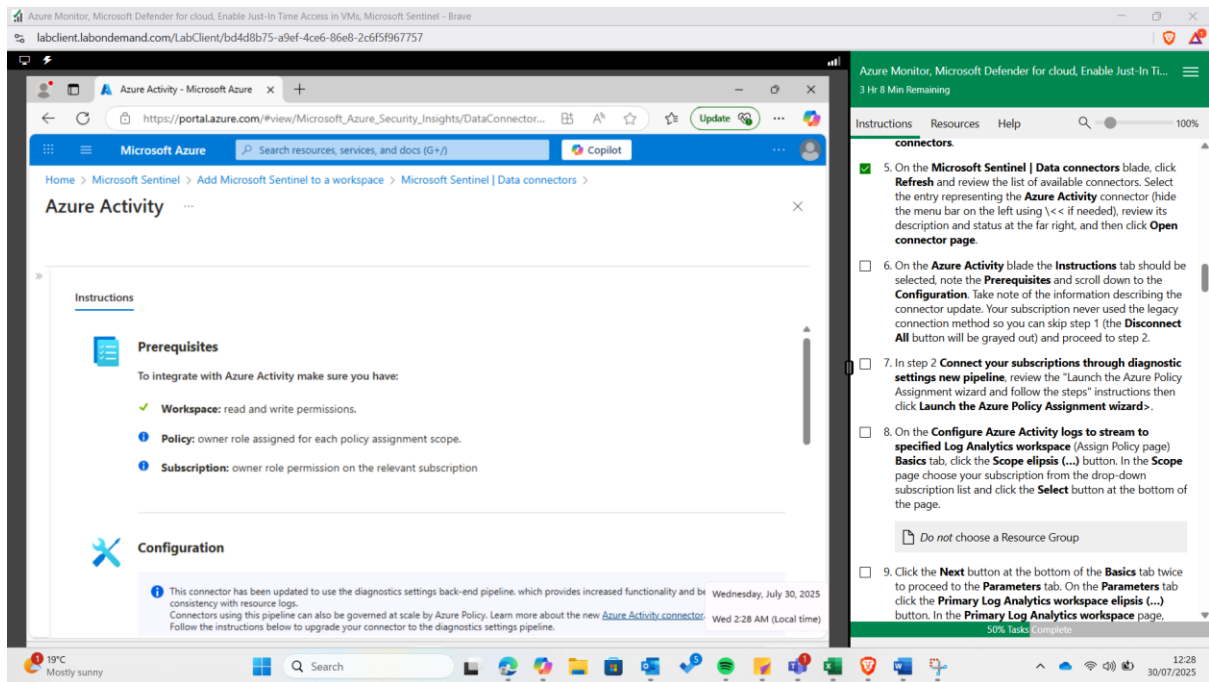
I typed **Azure** into the search bar, scrolled and selected the entry representing **Azure Activity**. Reviewed its description at the far right, and then clicked **Install**.



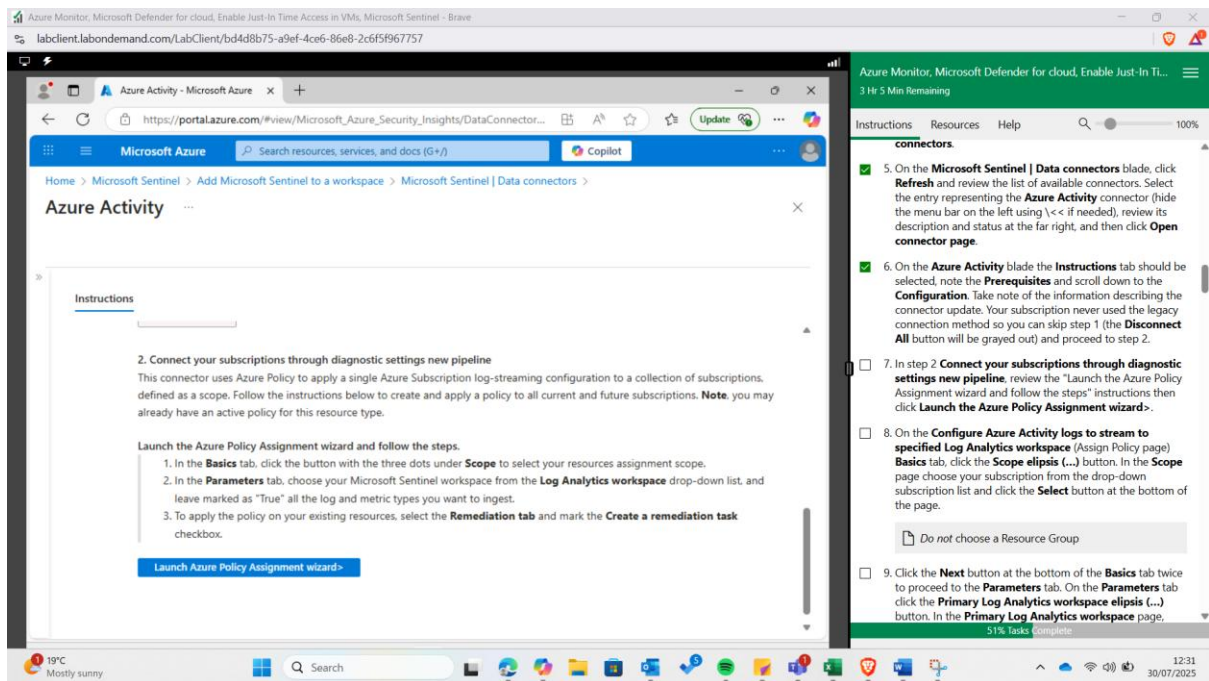
After receiving the Install success notification, I navigated to the left navigation panel, in the **Configuration** section, clicked **Data connectors**.

On the **Microsoft Sentinel | Data connectors** blade, I clicked **Refresh** and reviewed the list of available connectors. Selected the entry representing the **Azure Activity** connector (hide the menu bar on the left using << if needed), I then reviewed its description and status at the far right, and then clicked **Open connector page**.





On the configuration tab step 2, clicked on Launch Azure policy Assignment Wizard:



I then filled the basics tab according to the instructions above, without selecting a resource group:

Azure Monitor, Microsoft Defender for cloud, Enable Just-In-Time Access in VMs, Microsoft Sentinel - Brave

labclient.labondemand.com/LabClient/bd4d8b75-a9ef-4ce6-86e8-2c6f5967757

Microsoft Azure

Home > Microsoft Sentinel > Add Microsoft Sentinel to a workspace

Assign policy

Basics Parameters Remediation Non-compliance

Scope

Subscription: MOC Subscription-Iod50891741

Resource Group: Optionally choose a Resource Group

Exclusions: Optionally select resource

Resource selectors (Expand): Using resource selectors, you refine this assignment's ap targeting specific subsets of Expand to learn more.

Basics

Policy definition: Configure Azure Activity

Version (preview):

Overrides (Expand): Using overrides, you can d ne referenced versions of d

Previous Next Review + create

Select Cancel Clear All Selections

3 H 4 Min Remaining

Instructions Resources Help

6. On the **Azure Activity** blade the **Instructions** tab should be selected, note the **Prerequisites** and scroll down to the **Configuration**. Take note of the information describing the connector update. Your subscription never used the legacy connection method so you can skip step 1 (the **Disconnect All** button will be grayed out) and proceed to step 2.

7. In step 2 **Connect your subscriptions through diagnostic settings new pipeline**, review the "Launch the Azure Policy Assignment wizard and follow the steps" instructions then click **Launch the Azure Policy Assignment wizard**.

8. On the **Configure Azure Activity logs to stream to specified Log Analytics workspace** (Assign Policy page) **Basics** tab, click the **Scope elipsis (...)** button. In the **Scope** page choose your subscription from the drop-down subscription list and click the **Select** button at the bottom of the page.

Do not choose a Resource Group

9. Click the **Next** button at the bottom of the **Basics** tab twice to proceed to the **Parameters** tab. On the **Parameters** tab click the **Primary Log Analytics workspace elipsis (...)** button. In the **Primary Log Analytics workspace** page, make sure your subscription is selected and use the **workspaces** drop-down to select the Log Analytics workspace you are using for Sentinel. When done click the **Select** button at the bottom of the page.

10. Click the **Next** button at the bottom of the **Parameters** tab to proceed to the **Remediation** tab. On the **Remediation**

52% Tasks Complete

12:32 30/07/2025

Azure Monitor, Microsoft Defender for cloud, Enable Just-In-Time Access in VMs, Microsoft Sentinel - Brave

labclient.labondemand.com/LabClient/bd4d8b75-a9ef-4ce6-86e8-2c6f5967757

Microsoft Azure

Home > Microsoft Sentinel > Add Microsoft Sentinel to a workspace

Assign policy

Basics Parameters Remediation Non-compliance

Subscription: MOC Subscription-Iod50891741

workspaces: LAWS3403761

Primary Log Analytics workspace: Primary Log Analytics required.

Previous Next Select Cancel

Give feedback

2 H 56 Min Remaining

Instructions Resources Help

6. On the **Azure Activity** blade the **Instructions** tab should be selected, note the **Prerequisites** and scroll down to the **Configuration**. Take note of the information describing the connector update. Your subscription never used the legacy connection method so you can skip step 1 (the **Disconnect All** button will be grayed out) and proceed to step 2.

7. In step 2 **Connect your subscriptions through diagnostic settings new pipeline**, review the "Launch the Azure Policy Assignment wizard and follow the steps" instructions then click **Launch the Azure Policy Assignment wizard**.

8. On the **Configure Azure Activity logs to stream to specified Log Analytics workspace** (Assign Policy page) **Basics** tab, click the **Scope elipsis (...)** button. In the **Scope** page choose your subscription from the drop-down subscription list and click the **Select** button at the bottom of the page.

Do not choose a Resource Group

9. Click the **Next** button at the bottom of the **Basics** tab twice to proceed to the **Parameters** tab. On the **Parameters** tab click the **Primary Log Analytics workspace elipsis (...)** button. In the **Primary Log Analytics workspace** page, make sure your subscription is selected and use the **workspaces** drop-down to select the Log Analytics workspace you are using for Sentinel. When done click the **Select** button at the bottom of the page.

10. Click the **Next** button at the bottom of the **Parameters** tab to proceed to the **Remediation** tab. On the **Remediation**

52% Tasks Complete

12:40 30/07/2025

Azure Monitor, Microsoft Defender for cloud, Enable Just-In-Time Access in VMs, Microsoft Sentinel - Brave

labclient.labondemand.com/LabClient/bd4d8b75-a9ef-4ce6-86e8-2c6f5967757

Microsoft Azure

Home > Microsoft Sentinel > Add Microsoft Sentinel to a workspace > Microsoft Sentinel | Data connectors >

Azure Activity

Instructions

You don't have subscriptions using the legacy method, please move to step 2

Disconnect All

2. Connect your subscriptions through diagnostic settings new pipeline

This connector uses Azure Policy to apply a single Azure Subscription log-streaming configuration to a collection of subscriptions, defined as a scope. Follow the instructions below to create and apply a policy to all current and future subscriptions. **Note**, you may already have an active policy for this resource type.

Launch the Azure Policy Assignment wizard and follow the steps.

1. In the **Basics** tab, click the button with the three dots under **Scope** to select your resources assignment scope.
2. In the **Parameters** tab, choose your Microsoft Sentinel workspace from the **Log Analytics workspace** drop-down list, and leave marked as "True" all the log and metric types you want to ingest.
3. To apply the policy on your existing resources, select the **Remediation** tab and mark the **Create a remediation task** checkbox.

Notifications

More events in the activity log →Dismiss all

- Remediation task creation succeeded
Creating remediation task "09541aa360a34c939fba0528" was successful. a few seconds ago
- Role Assignments creation succeeded
All role assignments were created successfully. a few seconds ago
- Creating policy assignment succeeded
Creating policy assignment "Configure Azure Activity logs to stream to specified Log Analytics workspace" in "MOC Subscription-Iod50891741" was successful. Please note that the assignment takes around 5-15 minutes to take effect. Wednesday, July 30, 2025 Wed 2:49 AM (Local time)

22°C Mostly sunny

2 H 47 Min Remaining

Instructions Resources Help

10. Click the **Next** button at the bottom of the **Parameters** tab to proceed to the **Remediation** tab. On the **Remediation** tab select the **Create a remediation task** checkbox. This will enable the "Configure Azure Activity logs to stream to specified Log Analytics workspace" in the **Policy to remediate** drop-down. In the **System assigned identity location** drop-down, select the region (East US for example) you selected earlier for your Log Analytics workspace.

11. Click the **Next** button at the bottom of the **Remediation** tab to proceed to the **Non-compliance message** tab. Enter a Non-compliance message if you wish (this is optional) and click the **Review + Create** button at the bottom of the **Non-compliance message** tab.

12. Click the **Create** button. You should observe three succeeded status messages: **Creating policy assignment succeeded**, **Role Assignments creation succeeded**, and **Remediation task creation succeeded**.

You can check the Notifications, bell icon to verify the three successful tasks.

13. Verify that the **Azure Activity** pane displays the **Data received** graph (you might have to refresh the browser page).

It may take over 15 minutes before the Status shows "Connected" and the graph displays Data received.

Task 3: Create a rule that uses the Azure Activity data

56% Tasks Complete

12:49 30/07/2025

Verify that the **Azure Activity** pane displays the **Data received** graph. It took over 15 minutes before the Status showed "Connected" and the graph displays Data received

The screenshot shows the Microsoft Sentinel 'Data connectors' page. The 'Azure Activity' connector is listed with a status of 'Connected'. A graph titled 'Data received' shows a line chart with data points starting from July 24 to July 29. The graph shows a significant increase in data received around July 28 and 29. The status bar at the bottom indicates '56% Tasks Complete'.

Task 3: Create a rule that uses the Azure Activity data connector.

In this task, I reviewed and created a rule that uses the Azure Activity data connector.

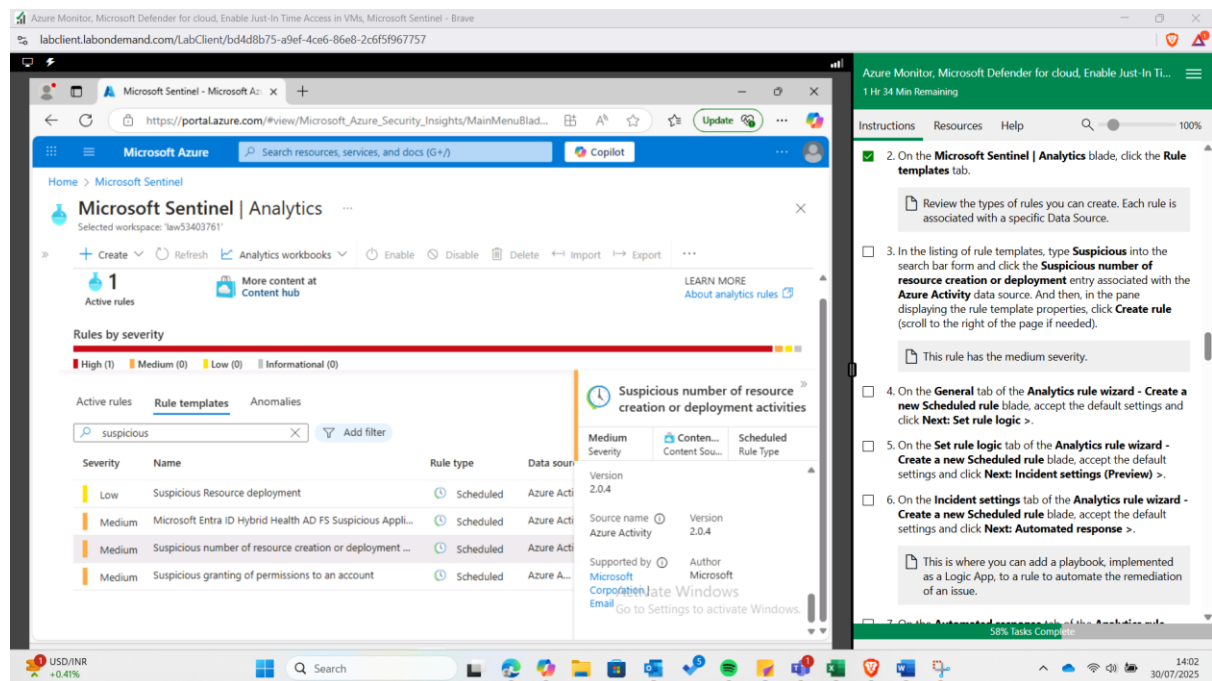
On the **Microsoft Sentinel | Configuration** blade, I clicked on **Analytics > Rule templates** tab

The screenshot shows the Microsoft Sentinel 'Analytics' page, specifically the 'Rule templates' tab. A table lists various rule templates with columns for Severity, Name, Rule type, Data sources, Tactics, and Techniques. The rules are categorized by severity: High (1), Medium (8), Low (8), and Informational (8). The 'Azure Activity' data source is highlighted for several rules.

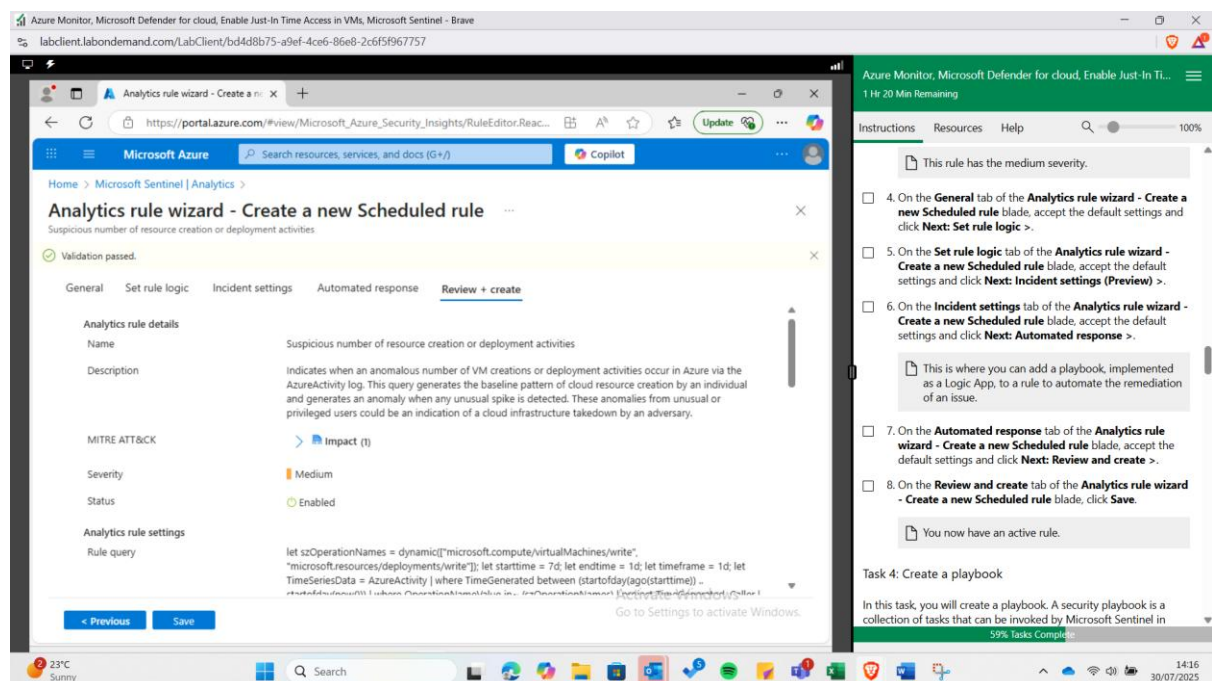
Severity	Name	Rule type	Data sources	Tactics	Techniques
Medium	NRT Microsoft ...	NRT	Azure Activity	Defense I	T1578
Low	New CloudShel...	Scheduled	Azure Activity	Executor	T1059
Low	Creation of exp...	Scheduled	Azure Activity	Defense I	T1578
Low	Rare subscrip...	Scheduled	Azure Activity	Azure Activity	T1093, T1149
Low	Suspicious Res...	Scheduled	Azure Activity	Impact	T1149

I reviewed the types of rules one can create. Each rule is associated with a specific Data Source.

In the listing of rule templates, I typed **Suspicious** into the search bar form and clicked the **Suspicious number of resource creation or deployment** entry associated with the **Azure Activity** data source. And then I right-clicked on it then pressed **Create rule**



Then I accepted all the default settings from the General tab till the end then I clicked save:



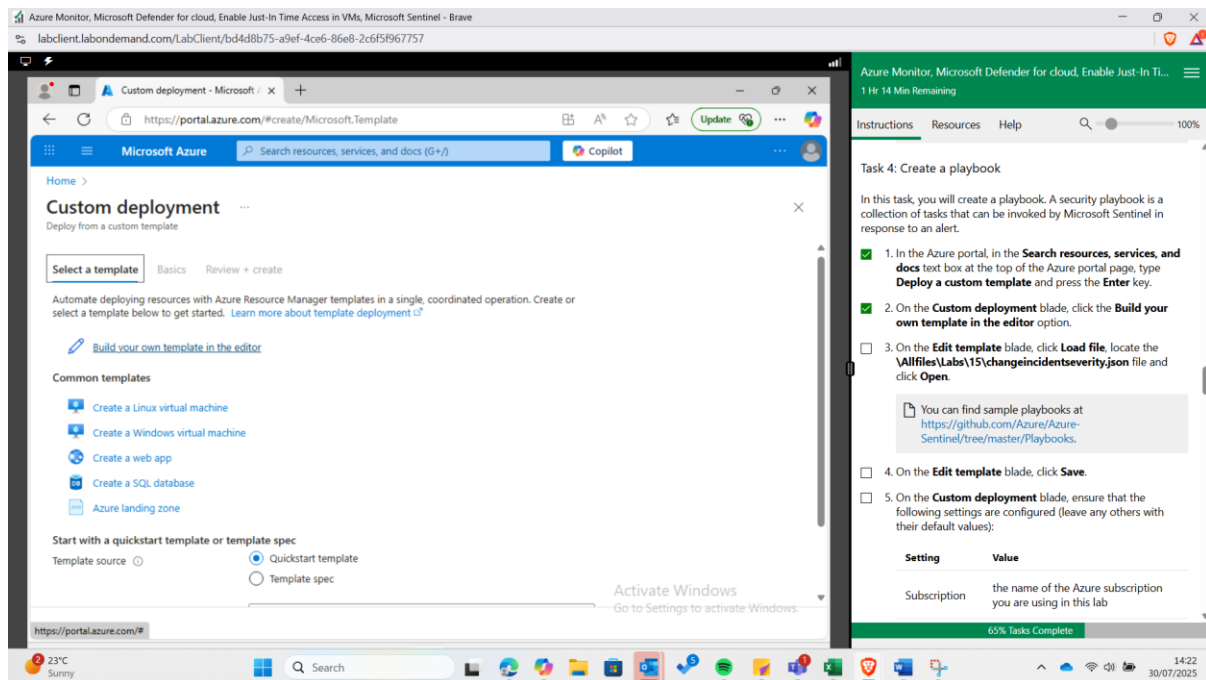
On the incident Settings tab, that is where one can add a playbook, implemented as a Logic App, to a rule to automate the remediation of an issue.

Once saved the rule is now active.

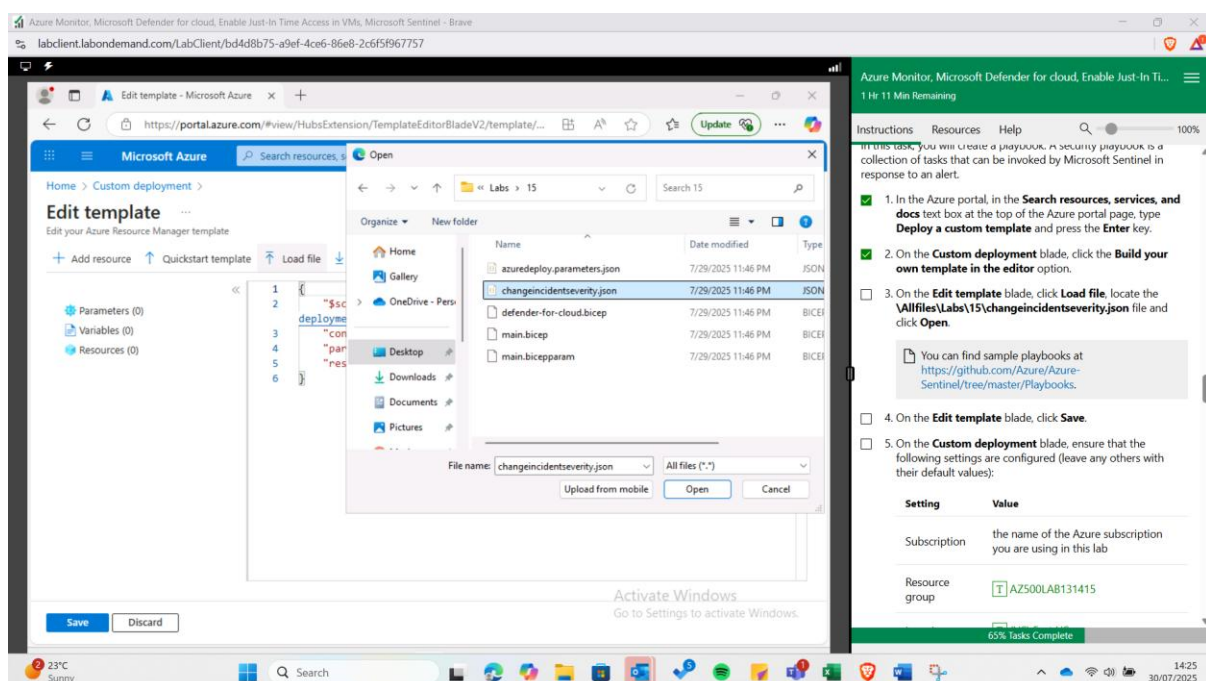
Task 4: Create a playbook

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, I typed **Deploy a custom template** and pressed the **Enter** key.

On the **Custom deployment** blade, I clicked the **Build your own template in the editor** option.

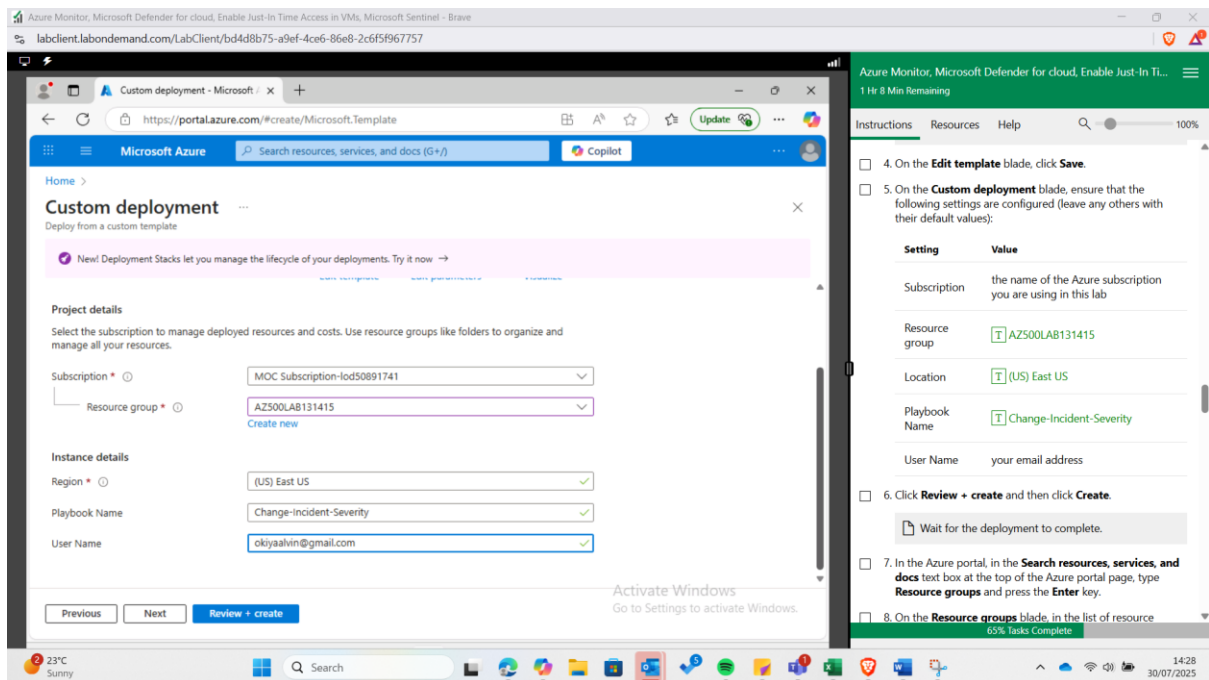


On the **Edit template** blade, I clicked **Load file**, located the **\Allfiles\Labs\15\changeincidentseverity.json** file and clicked **Open**

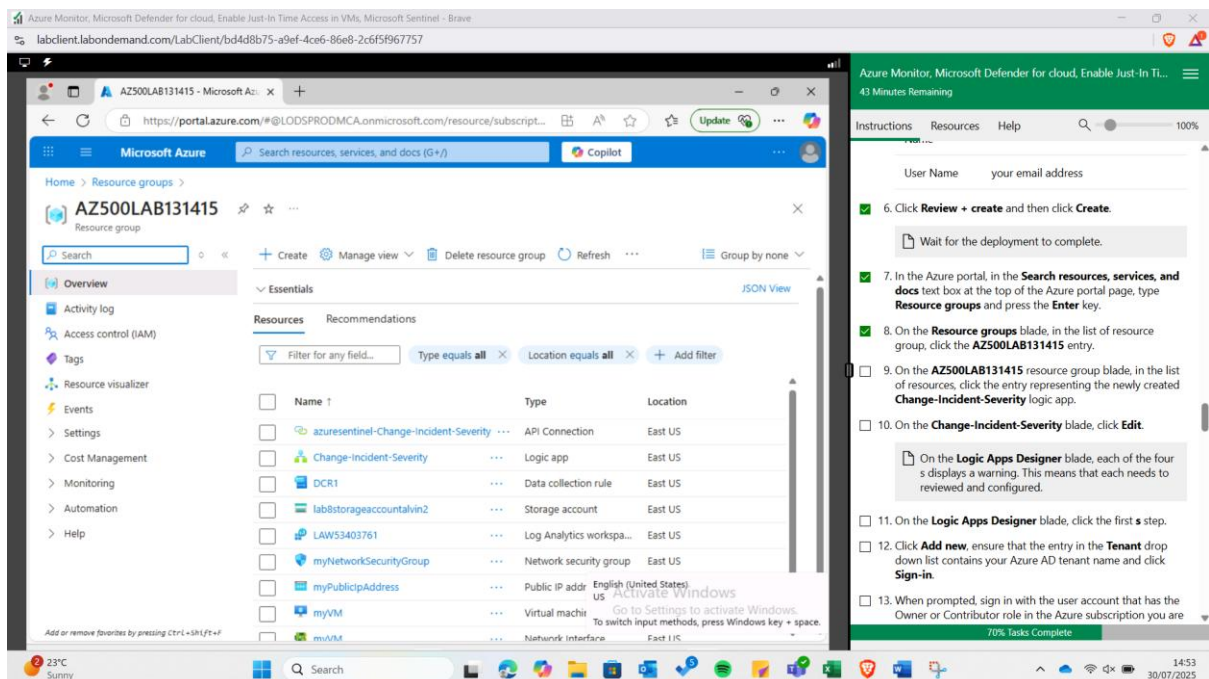


I then clicked save on the custom deployment blade.

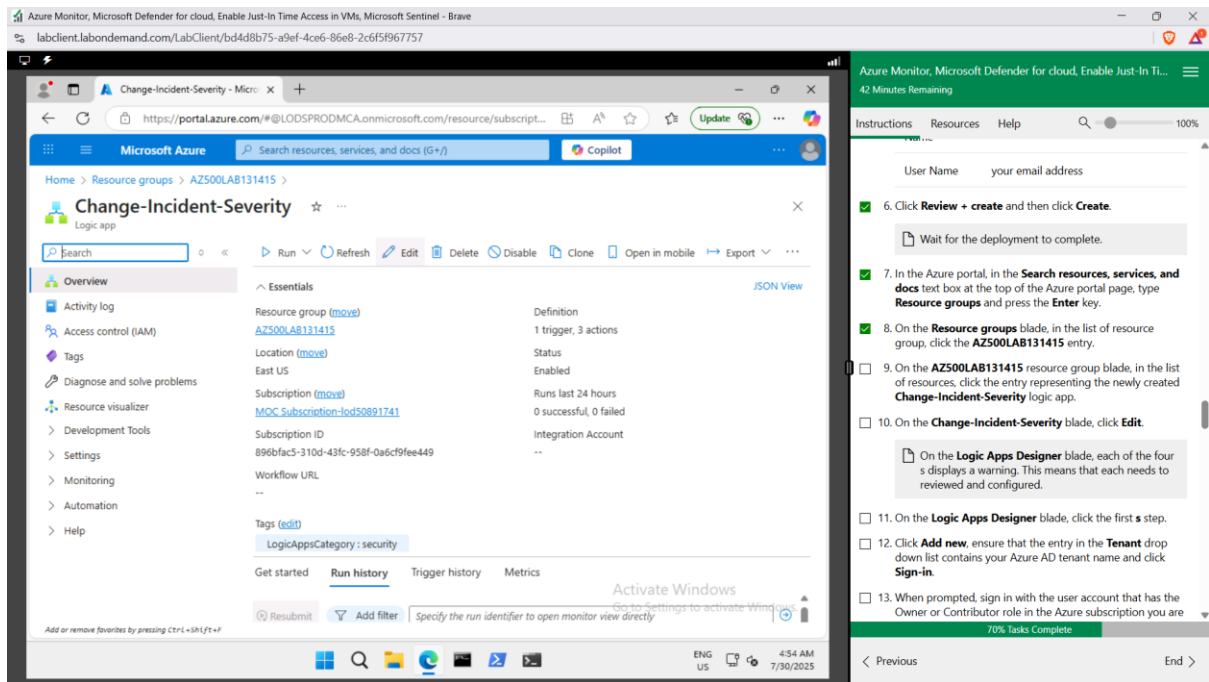
I then filled the basics tab as shown then clicked on **review + create** then **create**.



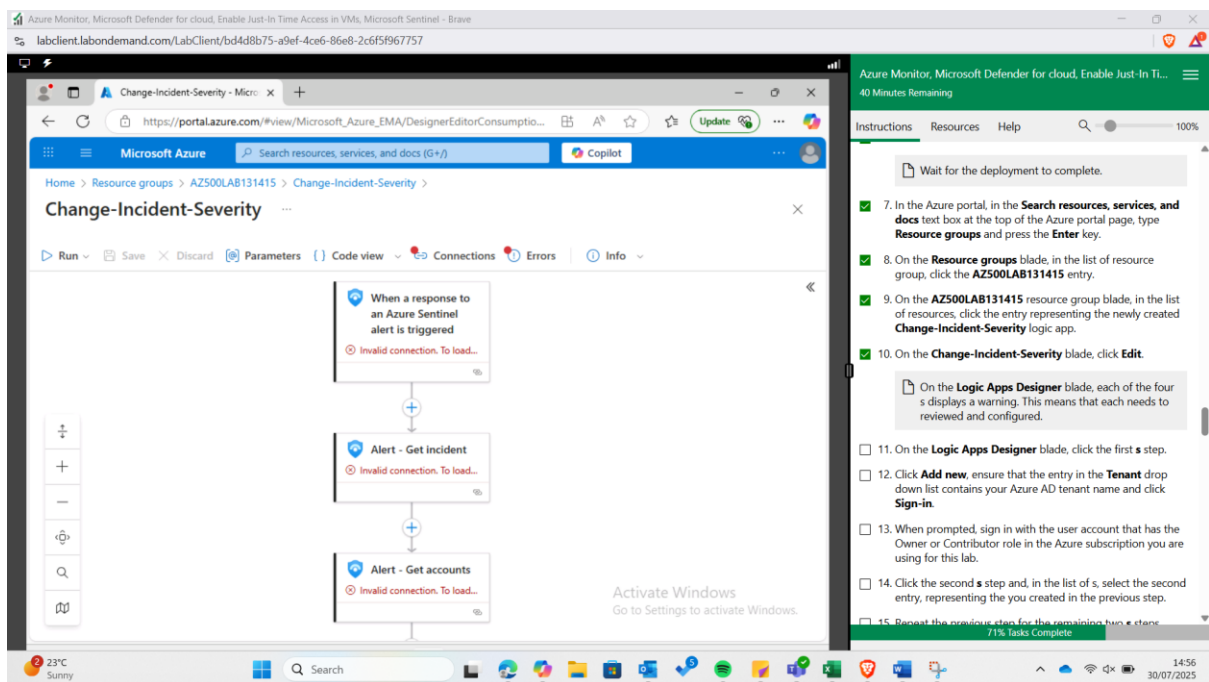
Next I navigated to the resource groups section of the portal and selected **AZ500LAB131415** resource group. On the **AZ500LAB131415** resource group blade, in the list of resources, I clicked the entry representing the newly created **Change-Incident-Severity** logic app.



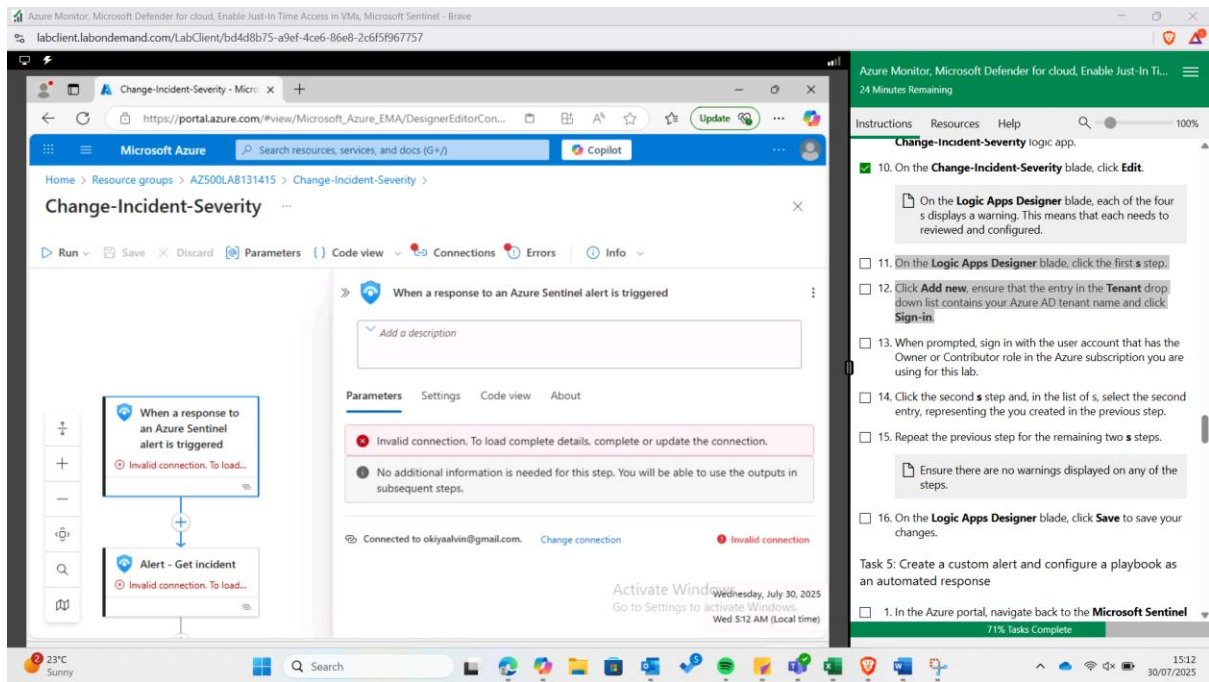
On the top blade in the Change-incident-severity blade, I clicked edit



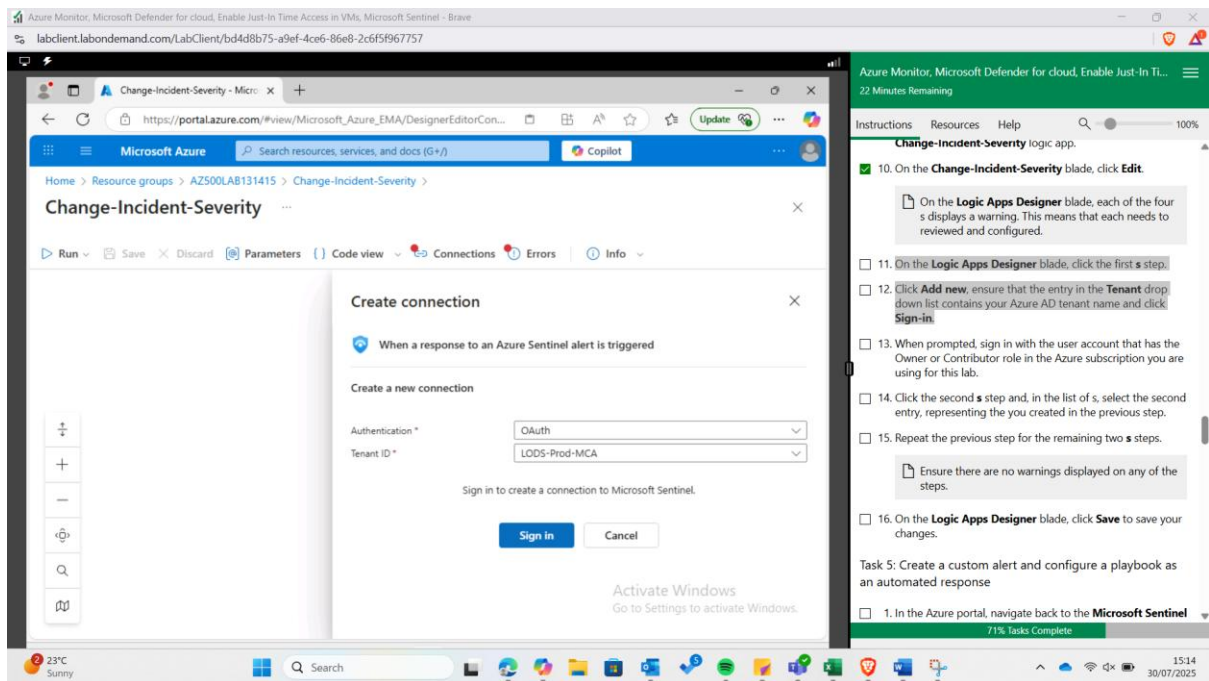
On the **Logic Apps Designer** blade, each of the four **s** displays a warning. This means that each needs to be reviewed and configured.



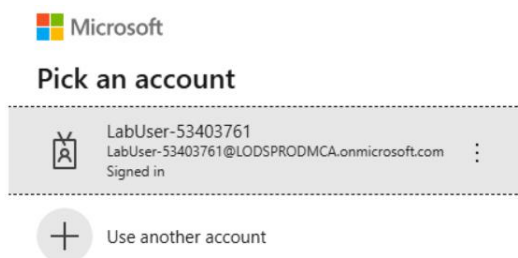
On the Logic Apps Designer blade, I clicked the first **s** step, and on the page below, I **clicked change connection**



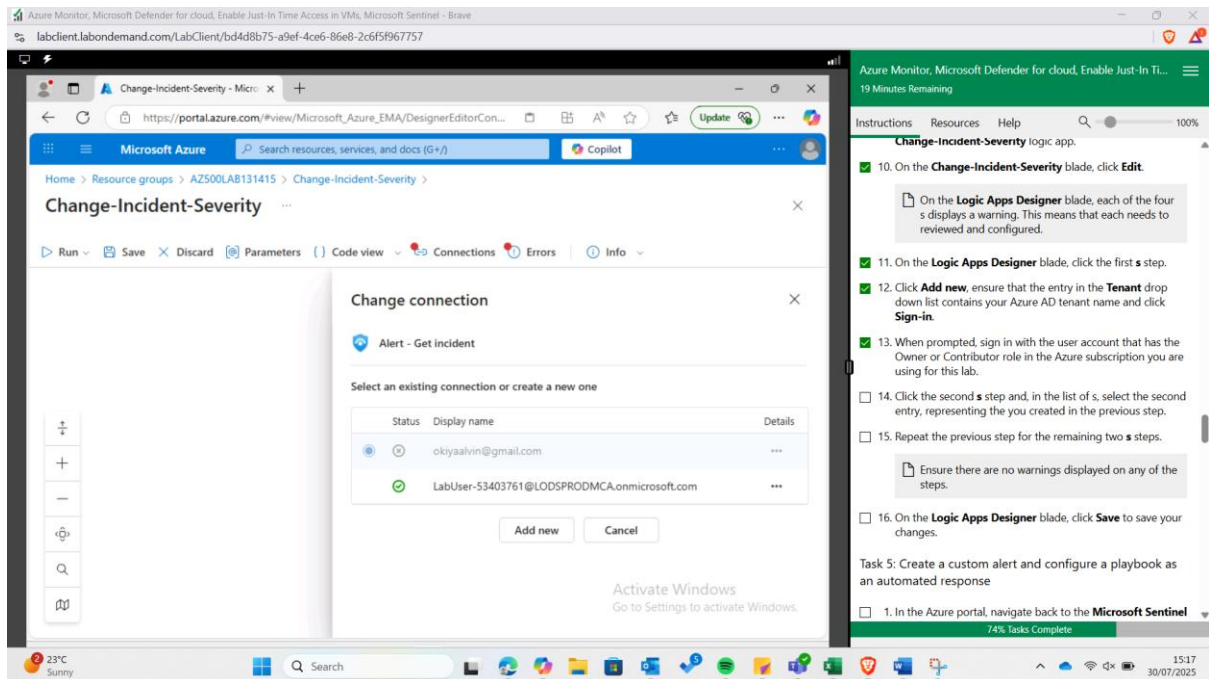
Then sign in:



Then signed in using the lab account I was using:



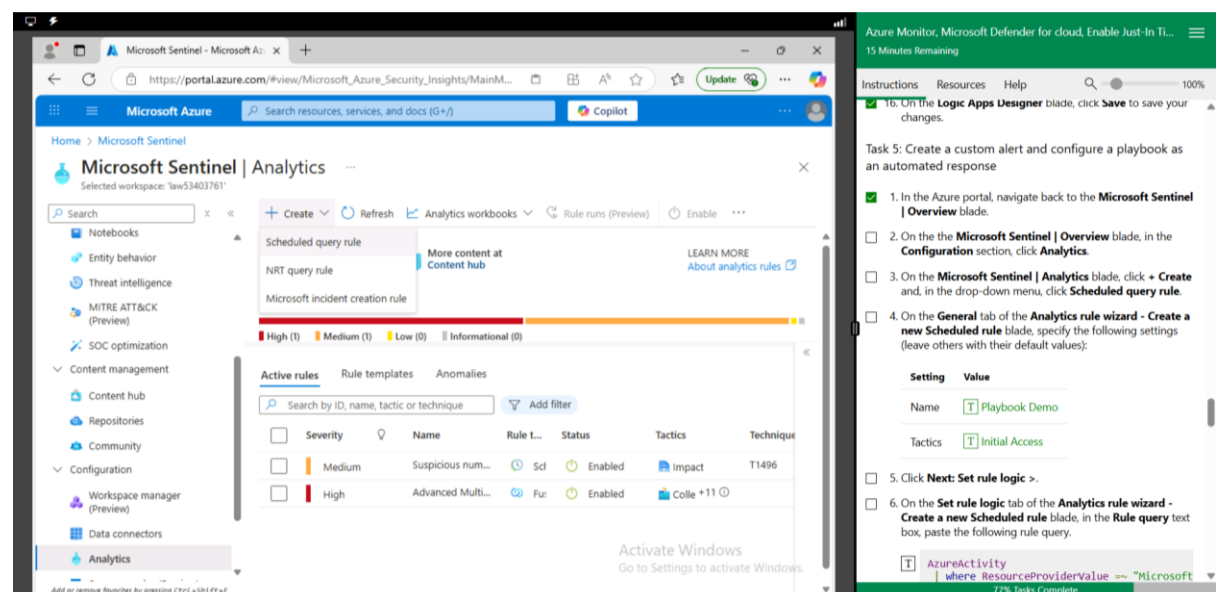
I then Clicked the second **s** step and, in the list of **s**, select the second entry, representing the you created in the previous step then clicked change settings, then selected the second option on the listed accounts, the one starting with “Lab user...”



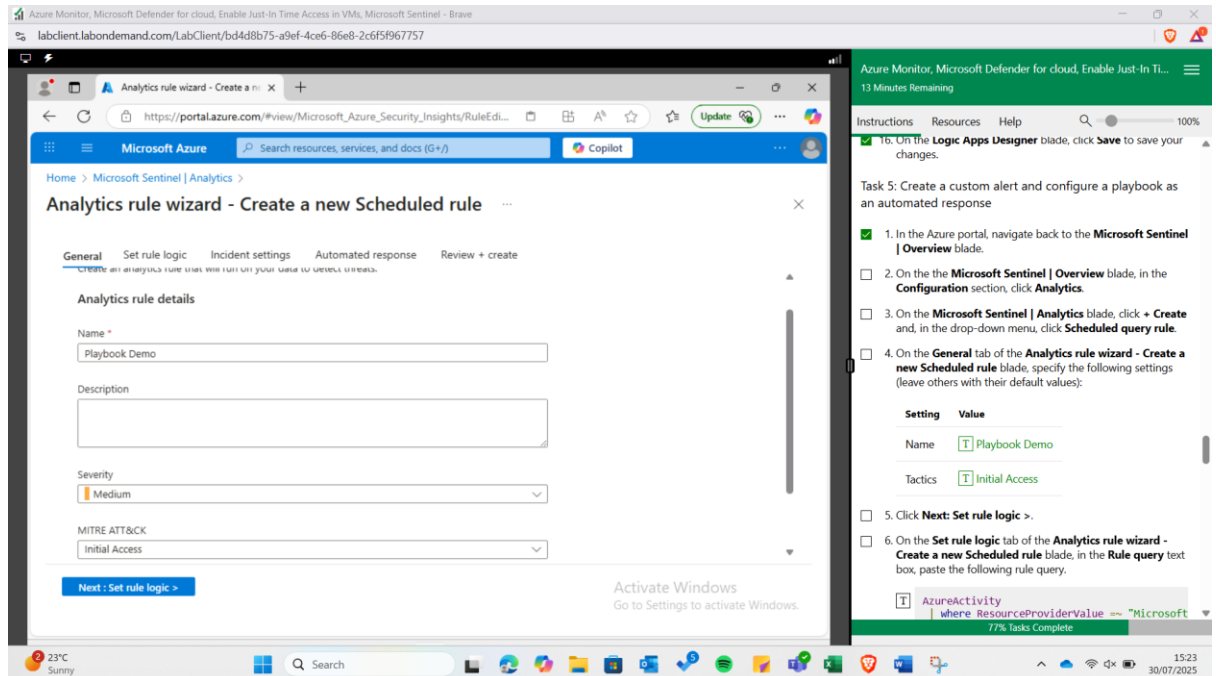
I repeated the previous step for the remaining two **s** steps, then clicked save at the top of the screen.

Task 5: Create a custom alert and configure the playbook as an automated response.

1. In the Azure portal, I navigated back to the **Microsoft Sentinel | Overview** blade.
2. On the the **Microsoft Sentinel | Overview** blade, in the **Configuration** section, I clicked **Analytics**.



- On the **Microsoft Sentinel | Analytics** blade, I clicked **+ Create** and, in the drop-down menu, I clicked **Scheduled query rule**.
- On the **General** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, I specified the following settings (left others with their default values):



- Then clicked **Next: Set rule logic**
- On the **Set rule logic** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, in the **Rule query** text box, paste the following rule query

AzureActivity

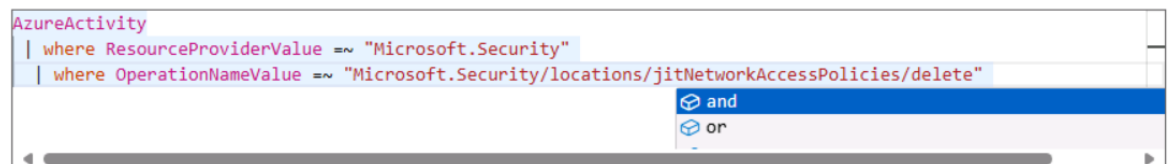
| where ResourceProviderValue =~ "Microsoft.Security"

| where OperationNameValue =~

"Microsoft.Security/locations/jitNetworkAccessPolicies/delete"

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.



[View query results >](#)

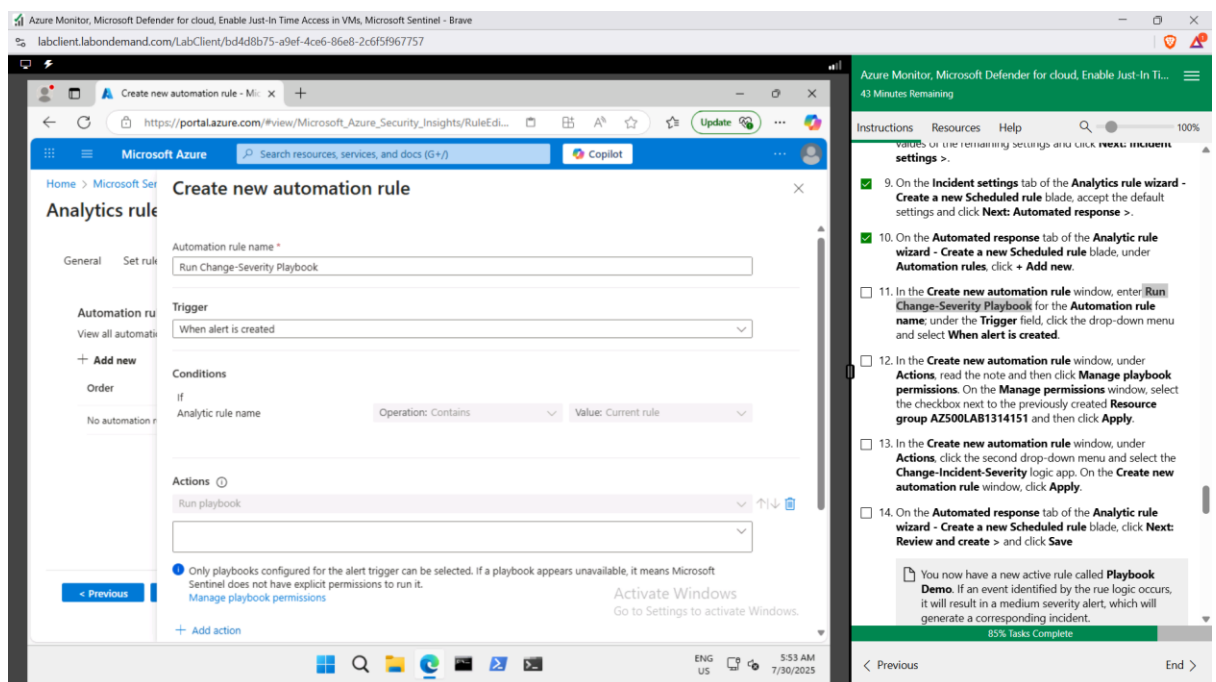
- Then on the **Set rule logic** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, in the **Query scheduling** section, I set the **Run query every** to **5 Minutes** and left other settings as default

Query scheduling

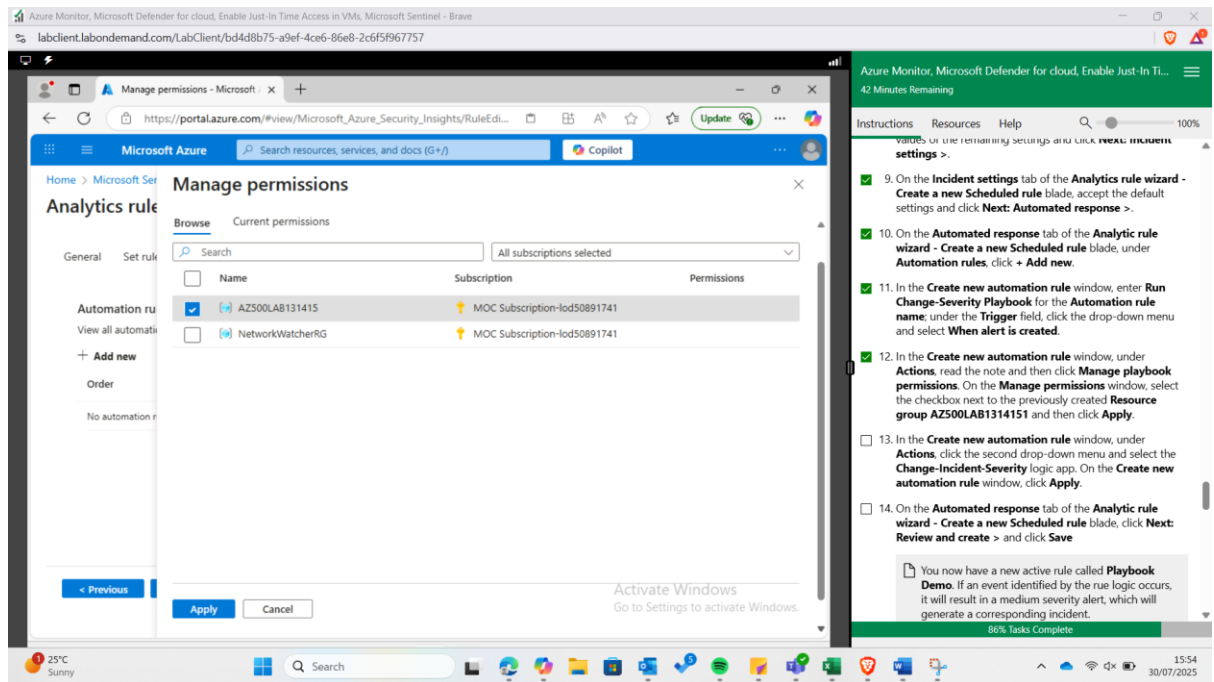
Run query every *

Lookup data from the last *

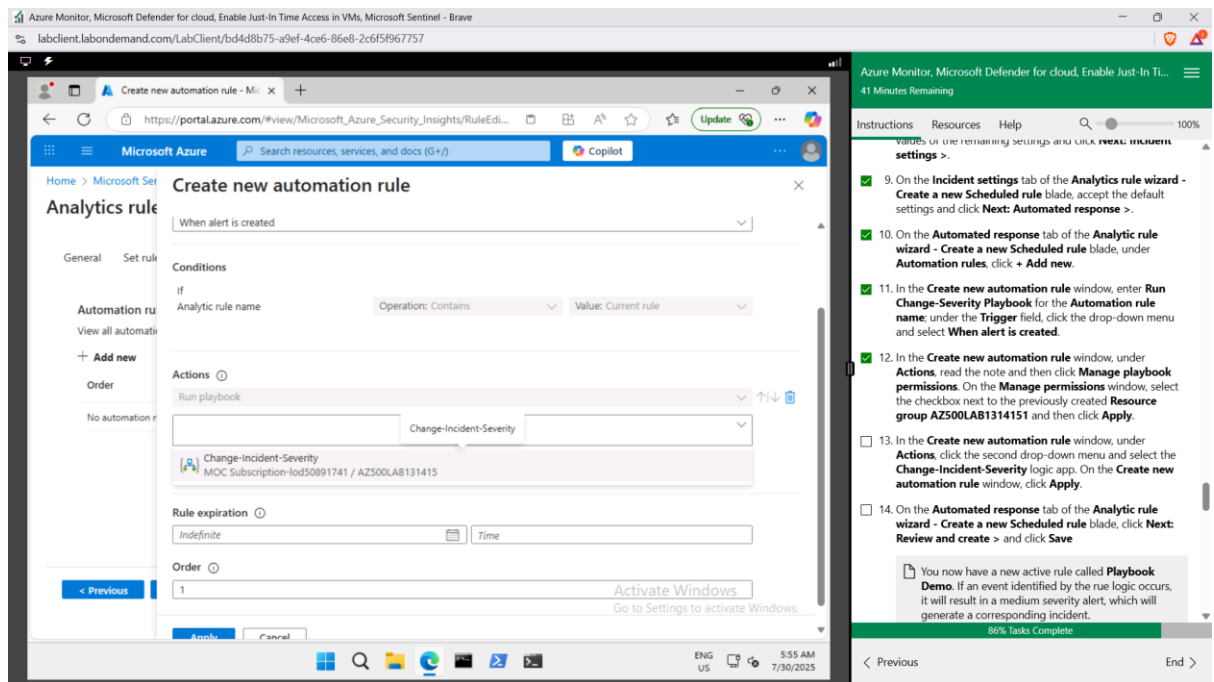
8. Then accepted other sections default settings till the **Automated response** tab of the **Analytic rule wizard - Create a new Scheduled rule** blade, where under **Automation rules** I clicked **+ Add new**.
9. In the Create new automation rule window, enter **Run Change-Severity Playbook** for the Automation rule name; under the **Trigger** field, I clicked the drop-down menu and selected **When alert is created**.



10. In the Create new automation rule window, under **Actions**, I clicked **Manage playbook permissions**. On the Manage permissions window, I selected the checkbox next to the previously created Resource group AZ500LAB1314151 and then clicked **Apply**.



11. In the Create new automation rule window, under Actions, I clicked the second drop-down menu and selected the **Change-Incident-Severity logic app**. On the Create new automation rule window, I clicked Apply.



12. I then clicked **Next: Review and create >** then **Save**

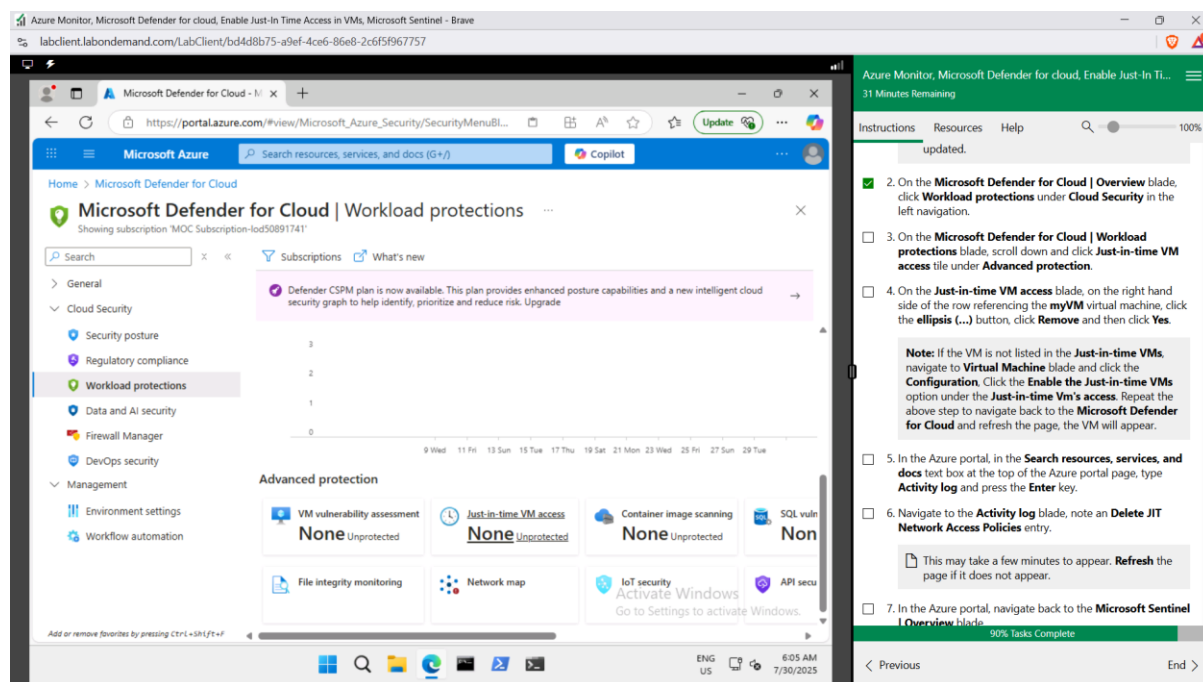
By doing this I had created a new active rule called **Playbook Demo**. If an event identified by the rule logic occurs, it will result in a medium severity alert, which will generate a corresponding incident.

Task 6: Invoke an incident and review the associated actions.

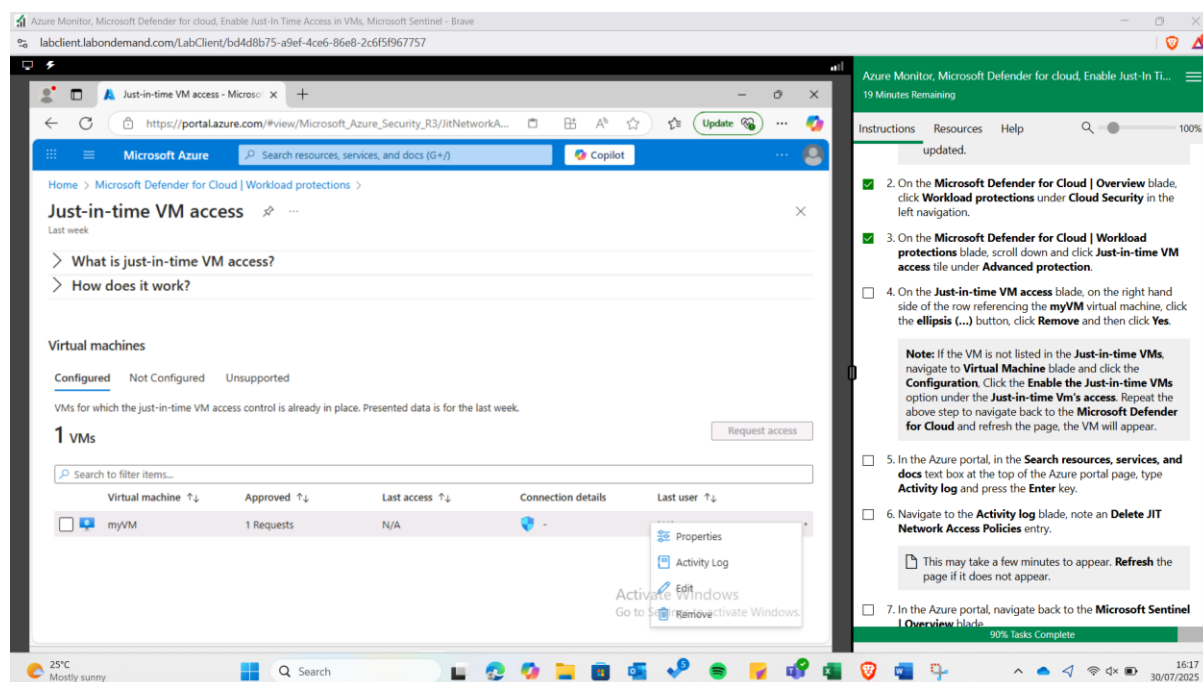
To do this task I navigated to the **Microsoft Defender for Cloud | Overview** blade.

Then under Cloud security clicked **Workload protections**.

On the **Microsoft Defender for Cloud | Workload protections** blade, I scrolled down and clicked **Just-in-time VM access** tile under **Advanced protection**.



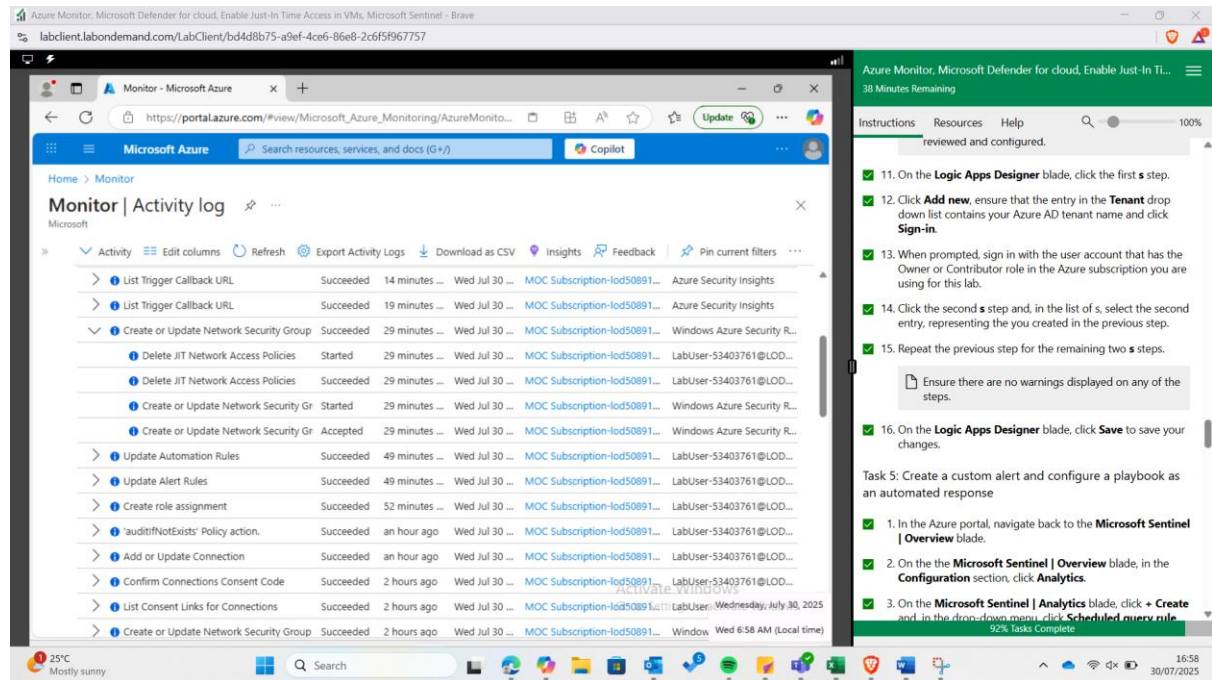
On the **Just-in-time VM access** blade, on the right hand side of the row referencing the **myVM** virtual machine, I clicked the **ellipsis (...)** button, then clicked **Remove** and then clicked **Yes**.



Note: If the VM is not listed in the **Just-in-time VMs**, navigate to **Virtual Machine** blade and click the **Configuration**, Click the **Enable the Just-in-time VMs** option under the **Just-in-time Vm's access**. Repeat the above step to navigate back to the **Microsoft Defender for Cloud** and refresh the page, the VM will appear.

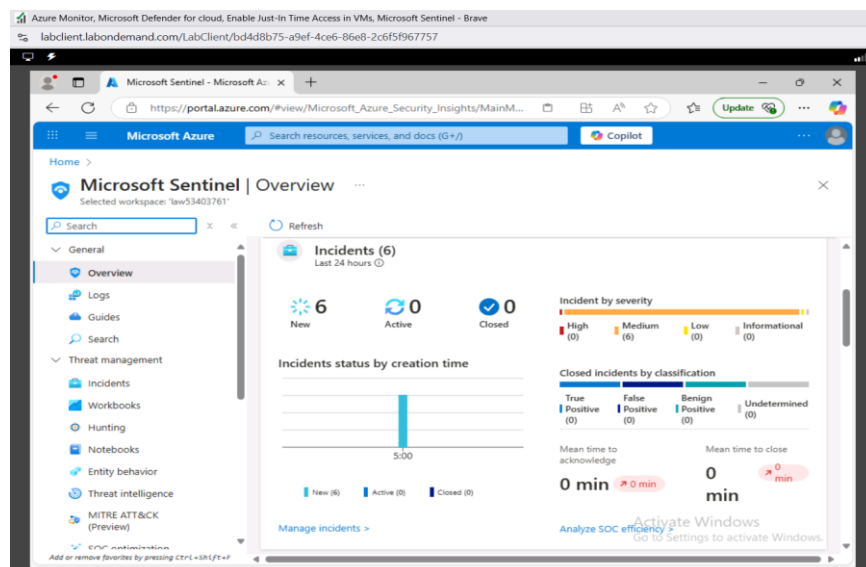
Next step, In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, I typed **Activity log** and pressed the **Enter** key.

I then navigated to the **Activity log** blade, noted a **Delete JIT Network Access Policies** entry



It was under create or update network security group.

To confirm this further I navigated to Microsoft Sentinel Overview blade where it could be seen from the dash board that there was an incident:



However, I navigated further to the Threat Management > Incidents section where the blade displayed an incident with both medium and high severity:

The screenshot shows the Microsoft Sentinel 'Incidents' blade. On the left, a navigation pane lists 'Overview', 'Logs', 'Guides', 'Search', 'Threat management', 'Incidents', 'Workbooks', 'Hunting', 'Notebooks', 'Entity behavior', 'Threat intelligence', and 'MITRE ATT&CK (Preview)'. The 'Incidents' section is active, displaying a summary with 9 Open incidents, 9 New incidents, and 0 Active incidents. A bar chart shows 'Open incidents by severity': High (1), Medium (8), Low (0), and Informational (0). Below this, a table lists incidents:

Severity	Incident number	Title	Alerts	Incident priority
High	9	Playbook Demo	1	Azure Sentinel
Medium	8	Playbook Demo	1	Azure Sentinel
Medium	7	Playbook Demo	1	Azure Sentinel

On the right, a sidebar contains instructions for creating a workspace and verifying incident severity. The bottom status bar shows 'Results: You have created a Microsoft Sentinel workspace, 96% Tasks Complete'.

Results: I have created a Microsoft Sentinel workspace, connected it to Azure Activity logs, created a playbook and custom alerts that are triggered in response to the removal of Just-in-time VM access policies, and verified that the configuration is valid.

Clean up

After completion of the task, I cleaned up resources using the command:

`Remove-AzResourceGroup -Name "AZ500LAB131415" -Force -AsJob`

The screenshot shows the Azure portal's 'Azure services' page. A Cloud Shell session is open at the bottom, displaying the command `Remove-AzResourceGroup -Name "AZ500LAB131415" -Force -AsJob` and its output. The output shows the job is running. On the right, a sidebar contains instructions for cleaning up resources. The bottom status bar shows 'Results: You have created a Microsoft Sentinel workspace, connected it to Azure Activity logs, created a playbook and custom alerts that are triggered in response to the removal of Just-in-time VM access policies, and verified that the configuration is valid. 99% Tasks Complete'.

Conclusion

This lab series significantly enhanced my understanding of how Microsoft Azure provides built-in capabilities to protect, monitor, and respond to threats across cloud environments. I learned how to deploy a Log Analytics workspace and configure data collection using the Azure Monitor Agent to centralize telemetry data. I also enabled Microsoft Defender for Servers, which adds advanced security controls and insights into virtual machines. Through Just-In-Time access, I was able to reduce the VM exposure to brute-force attacks by tightly controlling port access. Finally, with Microsoft Sentinel, I created a full incident detection and automation workflow using playbooks and custom alerts, showcasing how modern SIEM solutions can accelerate threat response.

These labs highlighted the power of integrating security tools within Azure to build a defense-in-depth strategy. Overall, this hands-on experience deepened my technical proficiency and preparedness to secure and monitor enterprise-grade cloud infrastructure.